Pico WiFi DMX User Manual

This manual explains how to use the browser-based DMX controller with the Pico firmware. It is written for daily operation: creating fixtures, controlling lights, saving scenes, building chasers, creating motion effects, and using GPIO buttons or ADC inputs.

The web interface is normally opened from XAMPP:

```
http://localhost/dmx/
```

The Pico itself is controlled over the network with its base URL, for example:

```
http://192.168.0.24/
```

Enter the Pico base URL once in any page. The browser stores it and shares it with the other pages. Across the UI, live Pico updates only happen while a Pico base URL is set; clearing the URL puts the page into browser-only editing.

This applies to every page that can talk to the Pico:

- **Fixture Controller** edits values locally, and only sends DMX changes when the Pico base URL is set.
- **Chaser** can edit steps locally without a URL; live step preview, Chase Playback, and Pico slot actions need a Pico base URL.
- **Motion FX** can edit effect recipes locally without a URL; browser preview, scene-to-Pico center updates, and Pico slot actions need a Pico base URL.
- **GPIO Control** can edit and save mappings locally without a URL; upload, readback, and status polling need a Pico base URL.
- **Benchmark** only runs when a Pico base URL is set.

The screenshots in this manual are generated by `scripts/update_user_manual.ps1`. Toolbox screenshots are captured as named page elements, so the manual can be regenerated after UI changes without hand-cropping images.

Main Pages

Page	Purpose
Fixture Controller	Fixture profiles, patching, live control, groups, scenes, default and blackout values
Chaser	Step-based chases with fade, loop modes, direction, pause/resume, and Pico slot upload
Motion FX	Continuous effects for pan/tilt pairs or scalar controls such as dimmer, zoom, iris, prism, and gobo
GPIO Control	Map Pico GPIO buttons and ADC inputs to lighting actions
Benchmark	Test Pico HTTP/DMX update performance

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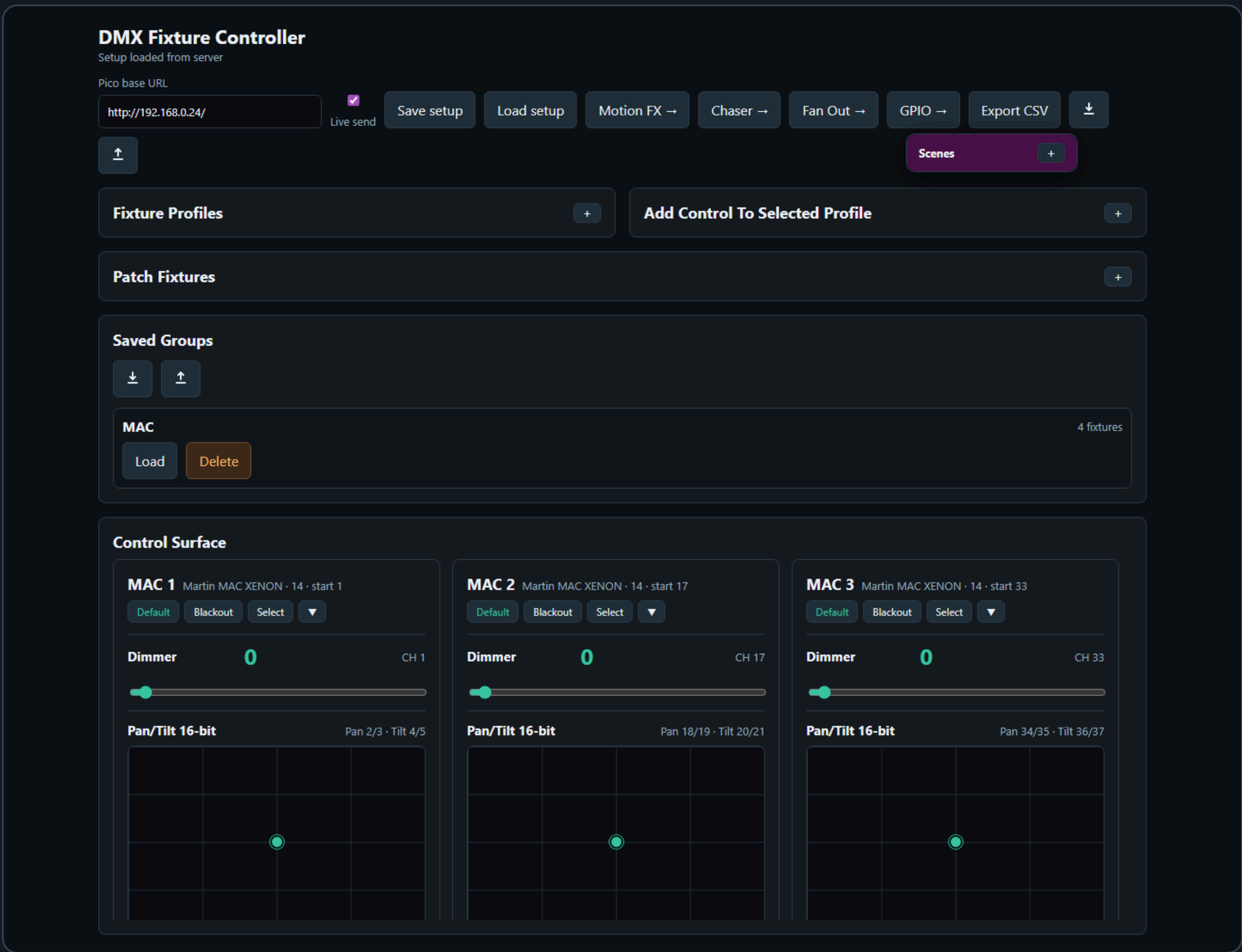
1. Fixture Controller



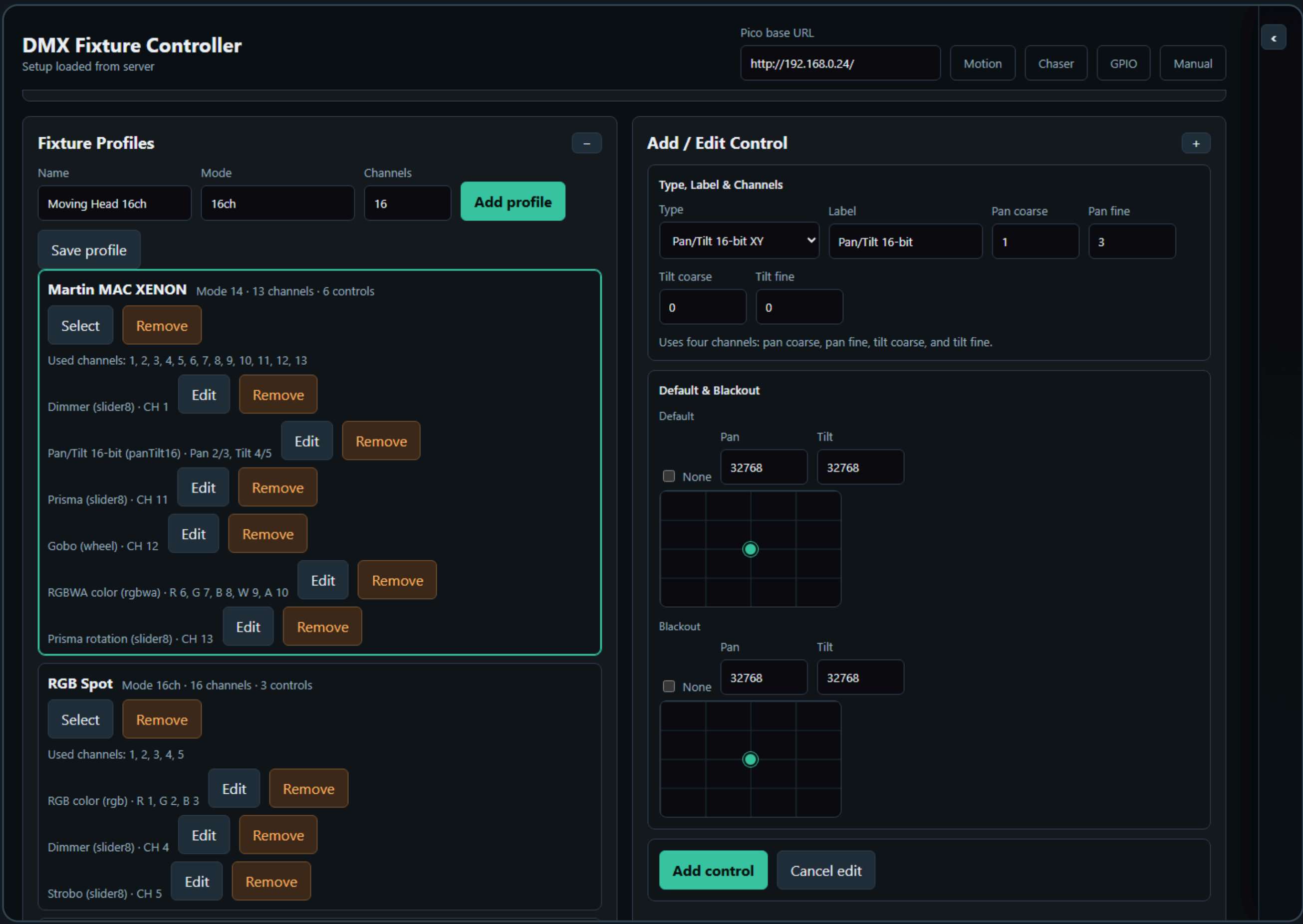
The Fixture Controller is the central page. Use it first when setting up a new show or fixture set.

Set the Pico Base URL

1. Open `http://localhost/dmx/`.

2. Enter the Pico base URL, for example `http://192.168.0.24/`.
3. Control changes are sent to the Pico when a Pico base URL is set. If the field is empty, the page edits locally only and does not send live DMX updates.
4. Fixture setup changes are autosaved to the XAMPP server. Use JSON export before large changes when you want an extra backup.

Create a Fixture Profile



A fixture profile describes the DMX personality of one fixture type.

1. Open **Fixture Profiles**.
2. Enter a profile name, mode name, and channel count.
3. Click **Add profile**.
4. Select the profile.
5. Use **Add / Edit Control** to add controls such as dimmer, pan/tilt, RGB, RGBW, RGBWA, wheel, or 16-bit slider.

Each control stores its own DMX channel mapping. For example, a moving head can have:

- Dimmer on channel 1
- Pan/Tilt 16-bit on channels 2/3 and 4/5
- RGBWA color on channels 6-10
- Gobo wheel on channel 12

To change an existing control, click **Edit** in the Fixture Profiles list. The **Add / Edit Control** card opens automatically and loads the selected control. After editing, click **Save control**. The **Fixture Profiles** collapse button also controls the Add / Edit Control card, so both profile editing areas can be hidden together.

Default and Blackout Values

Each control can store a **Default** value and a **Blackout** value.

Default values are useful for a normal starting look, for example:

- Dimmer open
- Pan/Tilt centered
- RGB white
- Gobo open

Blackout values are useful for quickly shutting down output, for example:

- Dimmer 0
- RGB black
- White and amber 0
- Pan/Tilt at a safe position if needed

For RGB, RGBW, and RGBWA controls, use the color picker for RGB. White and amber channels are edited separately when the fixture type has them.

Patch Fixtures

Patch fixtures after the profile is ready.

1. Open **Patch Fixtures**.
2. Enter a fixture name.
3. Select the fixture profile.
4. Enter the DMX start address.
5. Set **Count** to the number of fixtures you want to patch.
6. Click **Patch fixture**.

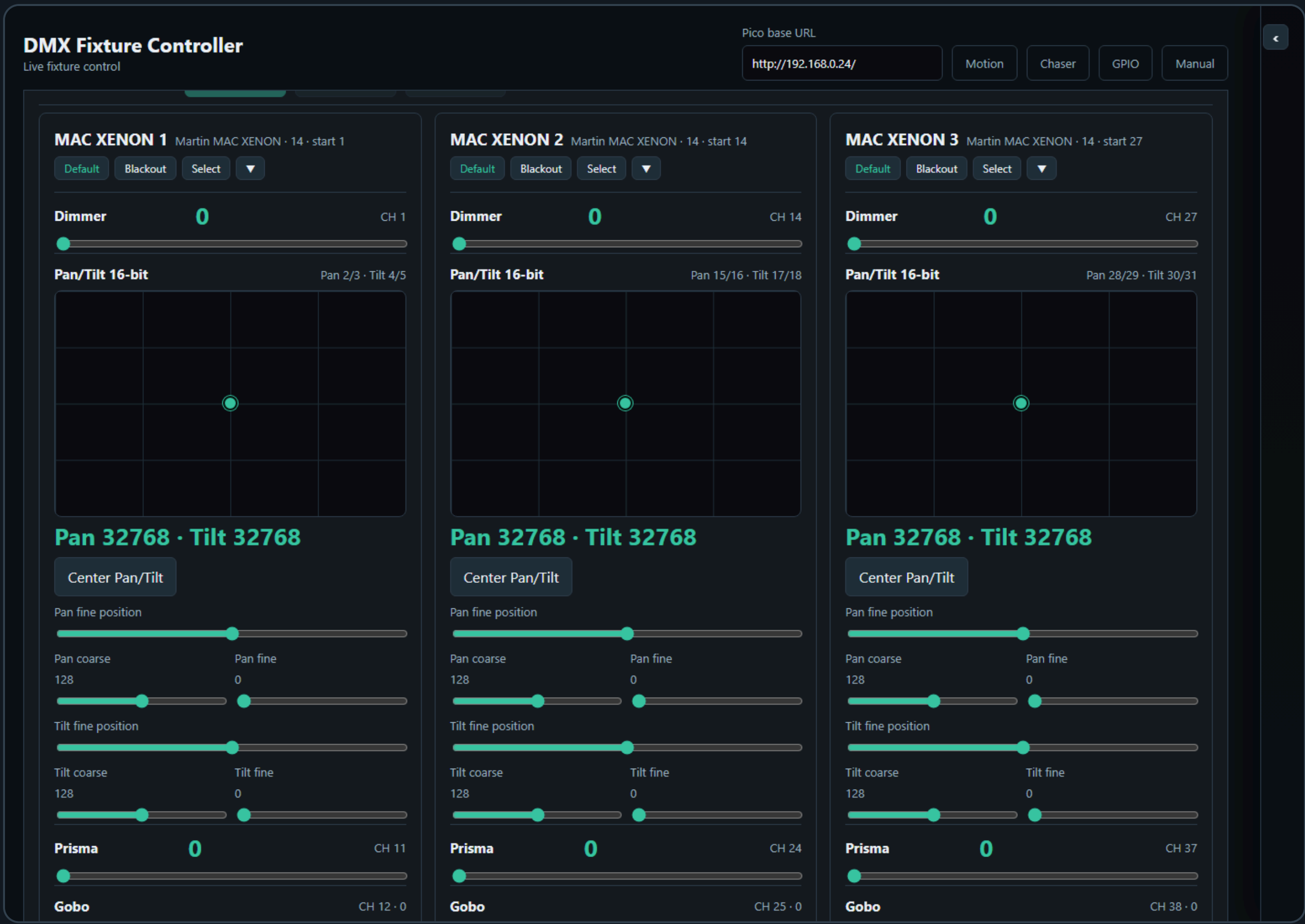
Use **Next free** to find the next available address after already patched fixtures.

When **Count** is greater than 1, the controller creates a numbered run of fixtures. For example, name `RGB Spot` with count `10` creates `RGB Spot 1` through `RGB Spot 10`. The first fixture uses the start address you entered. The following fixtures are spaced by the channel count of the selected fixture profile.

After a multi-fixture patch, the controller asks whether to create a Saved Group for the newly patched fixtures. The suggested group name is the patch name. Press **Cancel** to skip group creation, or enter a name to save the new fixtures as a group.

The patched fixture list is shown as rows of compact cards. Each row represents one consecutive run of the same fixture profile, so two separate MAC runs remain visually separate even when they use the same profile.

Live Fixture Control



The **Control Surface** shows one card per patched fixture.

Each fixture card contains the controls from its profile:

- Sliders for dimmer and simple channels
- XY pads for pan/tilt
- Color pickers and swatches for color controls
- Wheel buttons for indexed wheel values
- Coarse/fine sliders for 16-bit channels

Wheel / indexed controls require unique DMX option values. If two wheel options use the same DMX value, the Add / Edit Control card refuses to save the control. This prevents ambiguous wheel buttons where the UI could not know which option should be highlighted after recall or chase editing.

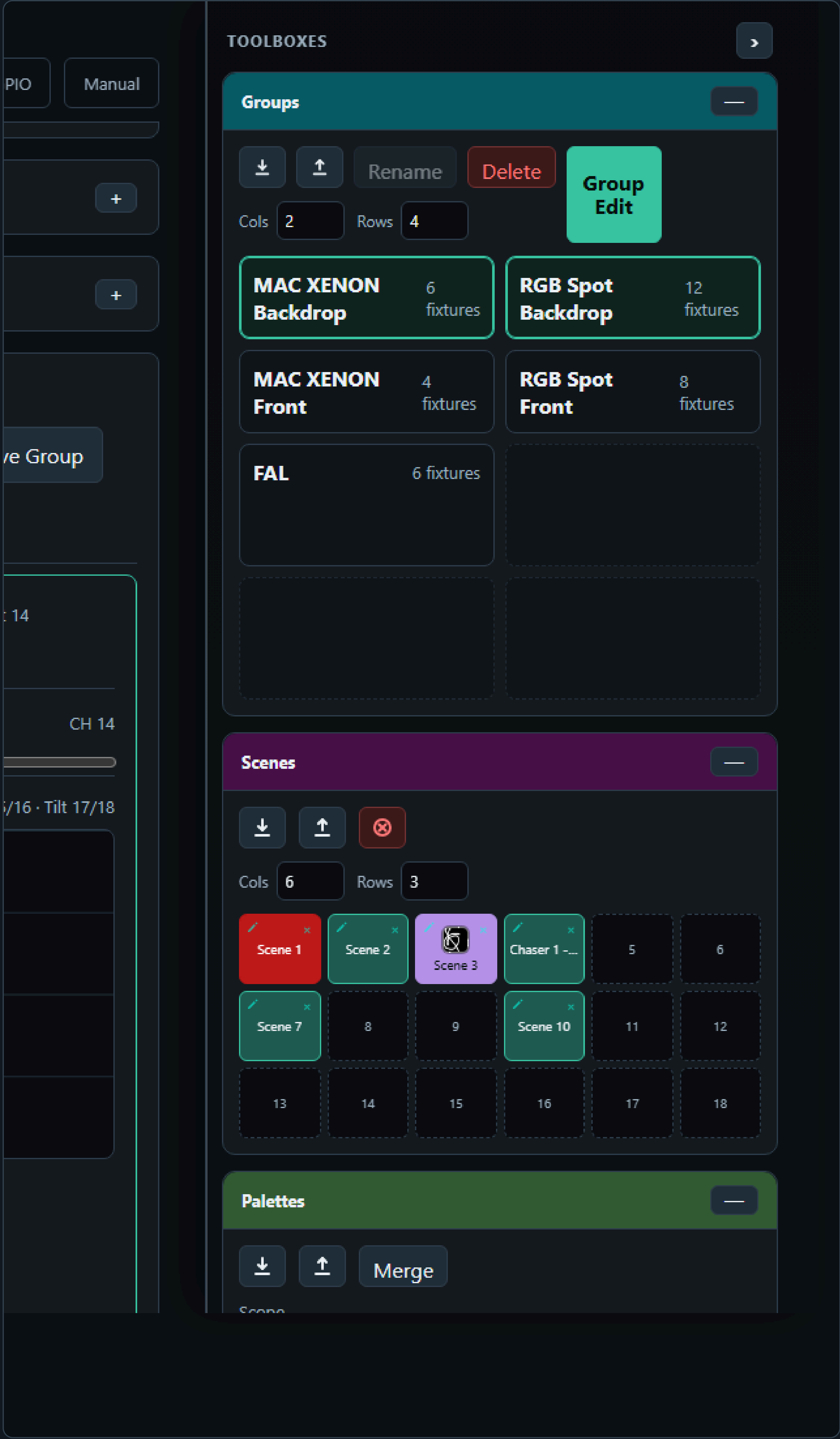
Use **Default** or **Blackout** on a fixture card to recall the stored values for that fixture only.

Use **Select** to include a fixture in group editing.

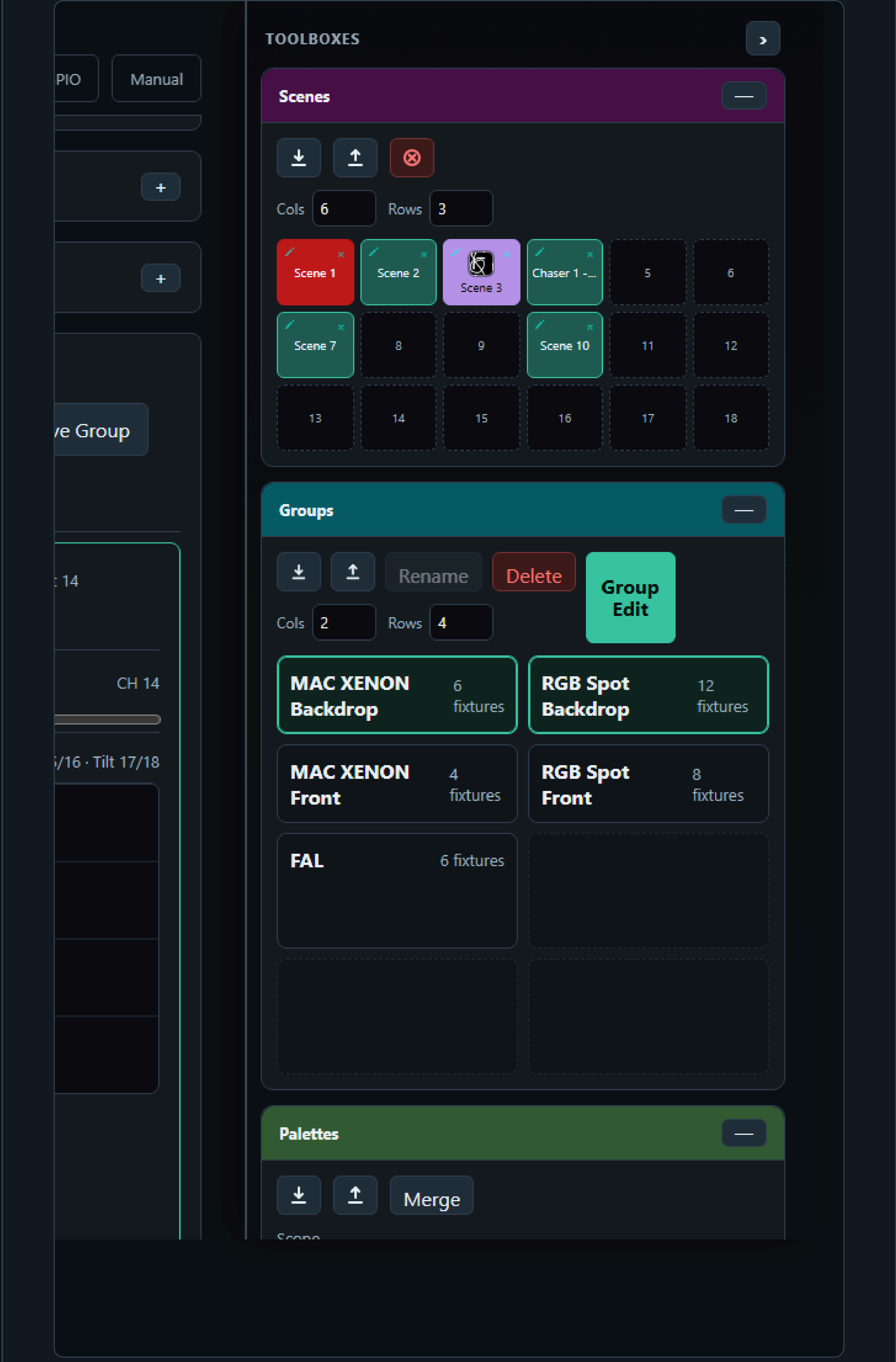
When you manually select fixture cards, the control surface stays visible so you can keep building or adjusting the selection. Saved Groups can also be selected from the Saved Groups card to filter the control surface.

Fixture Controller Toolboxes

The Fixture Controller uses four toolboxes in the shared right-side **Toolboxes** sidebar.


TOOLBOXES
PIO
Manual
ve Group
: 14
CH 14
5/16 · Tilt 17/18
Groups
Rename
Delete
Group Edit
Cols 2
Rows 4
MAC XENON Backdrop 6 fixtures
RGB Spot Backdrop 12 fixtures
MAC XENON Front 4 fixtures
RGB Spot Front 8 fixtures
FAL 6 fixtures
Scenes
Cols 6
Rows 3
Scene 1
Scene 2
Scene 3
Chaser 1 -...
5
6
Scene 7
8
9
Scene 10
11
12
13
14
15
16
17
18
Palettes
Merge
Scene

Groups stores fixture groups and shares the selected group filter with other toolbox pages. Use it to select fixtures quickly, rename one selected group, delete selected groups after confirmation, import/export group JSON, and open **Group Edit** when the current scope supports it.


PIO
Manual
ve Group
14
CH 14
/16 · Tilt 17/18
TOOLBOXES
Scenes
Cols 6
Rows 3
Scene 1
Scene 2
Scene 3
Chaser 1 -...
5
6
Scene 7
8
9
Scene 10
11
12
13
14
15
16
17
18
Groups
Rename
Delete
Group Edit
Cols 2
Rows 4
MAC XENON Backdrop
6 fixtures
RGB Spot Backdrop
12 fixtures
MAC XENON Front
4 fixtures
RGB Spot Front
8 fixtures
FAL
6 fixtures
Palettes
Merge

Scenes stores complete looks for the current working scope. Empty slots save, filled slots recall, the small `x` deletes, and the small top-left pencil icon edits that slot's background and optional drawing/upload.

PIO
Manual

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ve Group

: 14

CH 14

5/16 · Tilt 17/18

TOOLBOXES

Palettes

Merge

Scope

Position

Empty slot saves the selected scope. Filled slot recalls as an overlay.

Cols 4 Rows 4

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Scenes

Cols 6 Rows 3

Scene 1	Scene 2	Scene 3	Chaser 1 -...	5	6
Scene 7	8	9	Scene 10	11	12
13	14	15	16	17	18

Groups

Rename Delete Group Edit

Cols 2 Rows 4

Palettes stores reusable value fragments such as positions, colors, gobos, dimmer looks, or Fan Out results. Palette recall applies only the stored controls and leaves unrelated values untouched.

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Manual

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TOOLBOXES

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Fan Out

—

Affecting 18 fixtures in selected order

Control

Mode

Symmetric spread

Spread

Invert

0

Save

Recall

Snapshot

Apply

Clear

MAC XE... 0 → 0	MAC XE... 0 → 0	MAC XE... 0 → 0	MAC XE... 0 → 0	MAC XE... 0 → 0
MAC XE... 0 → 0	RGB Spot 1 128 → 128	RGB Spot 2 128 → 128	RGB Spot 3 128 → 128	RGB Spot 4 128 → 128
RGB Spot 5 128 → 128	RGB Spot 6 128 → 128	RGB Spot 7 128 → 128	RGB Spot 8 128 → 128	RGB Spot 9 128 → 128
RGB Spot... 128 → 128	RGB Spot... 128 → 128	RGB Spot... 128 → 128		

Palettes

—

Merge

Scope

Position

Empty slot saves the selected scope. Filled slot recalls as an overlay.

Cols 4 Rows 4

1	2	3	4

Fan Out shapes the currently selected fixtures or groups by spreading one compatible control through the selected order. It is documented in more detail below because its output can be saved as scenes or palettes.

Fan Out Toolbox

The Fixture Controller includes a **Fan Out** toolbox in the shared Toolboxes sidebar. It shapes the current controller values for the selected group or selected fixture cards, so the result can immediately be saved as a scene.

1. Select one or more Saved Groups, or select fixture cards manually.
2. In **Fan Out**, choose the control to shape, for example Dimmer, Pan, Tilt, or another compatible single-value control.
3. Click **Snapshot** to use the current controller values as the fan base.
4. Adjust **Spread**, or use **Start to end** mode with From/To offsets.
5. Use **Invert** when the spread direction should run through the selected fixtures in the opposite order.
6. The Control Surface updates continuously while you change the fan values.
7. The affected fixture controls are highlighted in the Control Surface.
8. Save the resulting look with the Scene Toolbox if you want to keep the actual DMX look.

The Fan Out toolbox only shows controls that are available on every fixture in the selected set. For pan/tilt controls, Pan and Tilt appear as separate fan targets. Applying a fan writes the calculated values into the controller just like moving the controls by hand. When a Pico base URL is set, the changed DMX values are also sent to the Pico.

Fan Out calculates from a stored base value for each fixture, control, and axis. **Snapshot** stores those base values. In **Symmetric spread**, the spread is added around the base values:

```
final value = snapshotted base value + spread offset
```

For example, with five fixtures and a spread of `100`, the offsets are `-50`, `-25`, `0`, `+25`, and `+50`. If every fixture has a base value of `128`, the result is `78`, `103`, `128`, `153`, and `178`. If the fixtures have different base values, each fixture still keeps its own base as the starting point.

Use **Save** in the Fan Out toolbox to store the fan setup itself: selected group or fixture IDs, selected control, mode, spread, and From/To offsets. Use **Recall** to restore that fan setup later. Recalling a Fan Out preset reapplies the fan to the controller values; it does not create a scene by itself. Use the Scene Toolbox when you want to store the resulting lighting look.

Palette Toolbox

The Fixture Controller also includes a **Palettes** toolbox. Palettes are reusable value fragments, while scenes are complete looks for their saved scope.

Use palettes for building blocks such as:

- Positions
- Colors
- Gobos
- Dimmer levels
- Fan Out results that you want to reuse as an overlay

The small top-left pencil icon on a filled palette slot opens the palette visual editor for that slot. The visual is independent from the palette scope: any palette can use a background color and an optional drawn/uploaded visual on top. The drawing canvas uses the selected background color, and the brush automatically switches between a light and dark stroke for readable contrast. Use **Default background** to restore the standard slot color, or **No icon** to keep only the colored button. Scope still decides which DMX values are saved; the visual is only a readable label shown in the palette slot grid and stored inside the palette JSON.

Palette save rules:

- If Fan Out is active, the palette saves only the Fan Out affected controls.
- If Fan Out is not active and fixtures are selected, the palette saves only the selected **Scope** for those fixtures.
- **Position** saves pan/tilt or position controls.
- **Color** saves RGB, RGBW, RGBWA, CMY, CMYK, and controls labeled as color-related.
- **Beam / Gobo** saves controls labeled as gobo, beam, prism, zoom, focus, iris, frost, strobe, or shutter.
- **Dimmer** saves controls labeled as dimmer, intensity, or shutter.
- **All controls** saves every control on the selected fixtures and should be used deliberately.
- If nothing is selected, the palette is not saved; select fixtures or apply Fan Out first.

Palette recall rules:

- Recalling a palette applies only the stored values.
- Unrelated controller values are left unchanged.
- The active group filter is cleared.
- The Control Surface is filtered to the fixtures involved in the palette.
- When a Pico base URL is set, only the recalled palette controls are sent to the Pico.

Filled palette slots recall palettes. Merging is done with the separate **Merge** button so recall and edit actions stay distinct.

Use **Merge** when you have a current fixture selection or active Fan Out result:

- The controller asks which saved palette slot should receive the current scoped values.
- If the current scope matches the saved palette scope, the values are merged and the palette keeps its scope.
- This allows one **Color** palette to contain RGB controls, color wheels, and RGBWA controls from different fixture types while still remaining a Color palette.
- If the current scope differs from the saved palette scope, the controller asks whether to change the saved palette to **All controls** and merge.
- If you cancel a merge prompt, the palette is not changed.

2. Scenes And Palettes

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TOOLBOXES

Scenes

Cols 6 Rows 3

Scene 1	Scene 2	Scene 3	Chaser 1 -...	5	6
Scene 7	8	9	Scene 10	11	12
13	14	15	16	17	18

Groups

Rename Delete **Group Edit**

Cols 2 Rows 4

MAC XENON Backdrop 6 fixtures	**RGB Spot Backdrop** 12 fixtures
MAC XENON Front 4 fixtures	**RGB Spot Front** 8 fixtures
FAL 6 fixtures	

Palettes

Merge

Scope

The Scene Toolbox is available on the Fixture Controller page.

Use scenes to store fixture/control looks. A scene stores controller values by fixture/control key, not a raw 512-channel DMX dump.

The small top-left pencil icon on a filled scene slot opens the same visual editor used by palettes. A scene can have a background color and an optional drawn/uploaded visual in its slot. The canvas background follows the selected background color, and the brush color is calculated for contrast against it. **Default background** restores the standard slot color, and **No icon** removes the drawn/uploaded image. This visual is only a label for finding the scene quickly; scene save and recall still use the stored fixture values.

1. Set your desired fixture values.
2. Click an empty scene slot.
3. Enter a scene name.
4. Click a filled slot once to recall it.
5. Use the small `x` on a filled slot to delete it.

Scene Save Rules

Scene saving uses the current working scope:

- If Fan Out is active, the scene saves only the fixtures affected by Fan Out.
- For each Fan Out fixture, the scene saves all controls of that fixture, not only the fanned control. This keeps related values such as Tilt stable when saving a Pan fan.
- If Fan Out is not active and fixtures are selected, the scene saves only the selected fixtures.
- Selected Saved Groups count as selected fixtures, so a group-filtered scene saves only the fixtures in the selected groups.
- If a scene was recalled and the control surface is filtered to that scene's involved fixtures, saving again uses those involved fixtures.
- If nothing is selected, the controller asks before saving all patched fixtures.
- If you cancel that prompt, no scene is saved.

This means a scene is normally scoped to the fixtures you are actually working with. It does not silently save unrelated fixtures unless you explicitly confirm the all-fixtures fallback.

Scene Recall Rules

Scene recall loads the stored values back into the Fixture Controller, redraws the fixture cards, and updates the live-value snapshot used by the Chaser page.

When a Pico base URL is set, scene recall also sends the recalled DMX values to the Pico in one batch. To work browser-only, clear the Pico base URL before recalling or editing looks.

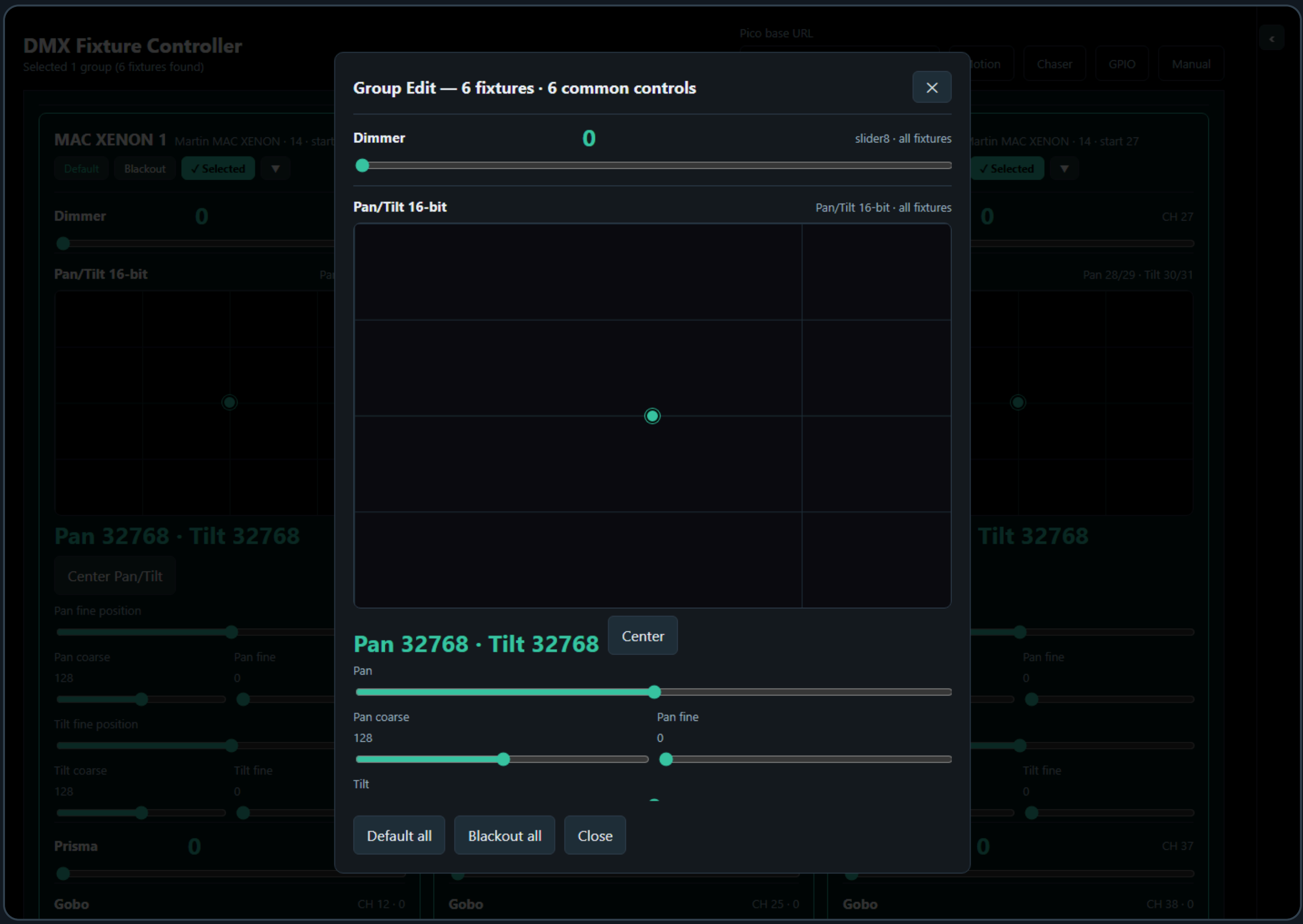
Recalling a scene clears the active group selection and shared group filter. It then selects the fixtures involved in the recalled scene and filters the Control Surface to only those fixtures. This makes the recalled scene visible without leaving an old group filter active, and without showing unrelated fixtures.

If a saved scene was created before fixtures were repatched, the controller tries to remap old fixture IDs to the current patch by matching the same fixture profile in DMX start-address order. This lets older scenes continue to recall after a fixture run was recreated, as long as the fixture profiles and control IDs still match.

The red **Clear all DMX channels** button clears controller values. When a Pico base URL is set, it also calls `/dmx/clear` on the Pico. This clears both live DMX output and the motion base buffer.

The Scene Toolbox sits in the shared **Toolboxes** sidebar. Its slot grid follows the configured rows and columns, and the tile size expands to the available sidebar width while keeping the old minimum slot size.

3. Groups

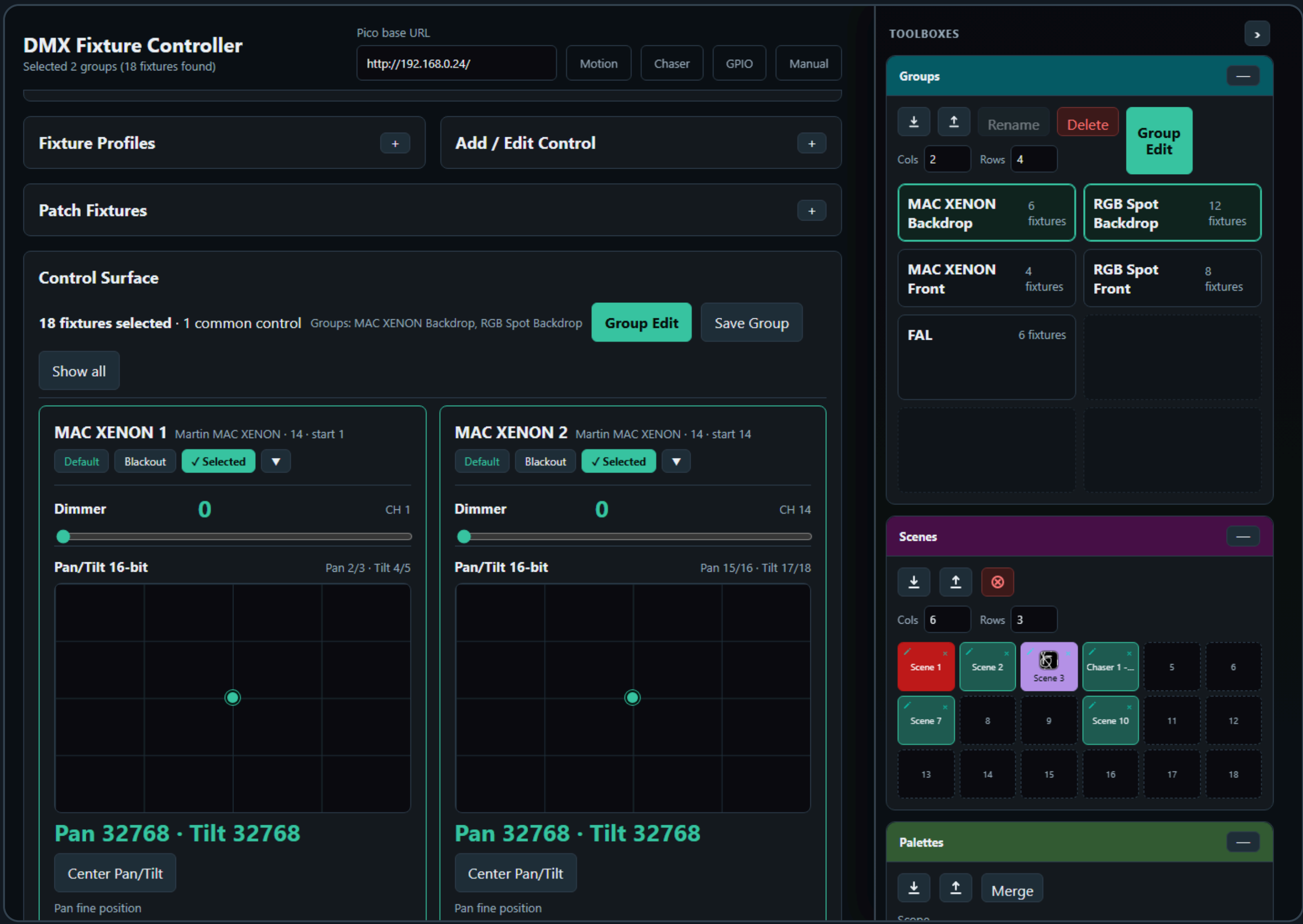


Groups let you control multiple fixtures at once.

Create a Group

1. Select two or more fixture cards.
2. Click **Save Group**.
3. Enter a group name.
4. The group appears in **Saved Groups**.

Saved Group Selection



The Saved Groups card shows groups in a compact matrix. Each group has four actions:

- **Select** adds the group to the active group filter.
- **Deselect** removes only that group from the active filter.
- **Rename** changes the saved group name.
- **Delete** removes the saved group.

More than one saved group can be selected at the same time. The control surface shows the union of all fixtures from the selected groups. This makes it possible to work with several fixture groups together without editing the group definitions.

The group bar above the control surface shows how many fixtures are selected and which saved groups are active. Use **Show all** to clear the saved-group filter and return to the full fixture list.

Group selection is shared across toolbox pages that use the Groups toolbox. It is a working filter: selecting groups limits the visible fixtures for controller editing, Fan Out targeting, and Chaser participating-control setup. Recalling a scene, editing a saved chase step, or loading a saved chase clears the group filter because the recalled data itself becomes the source of truth.

Edit a Group

1. Load a saved group or select multiple fixtures.
2. Click **Group Edit**.
3. Adjust common controls in the modal.

The Group Edit modal only shows controls that exist on all selected fixtures. This prevents accidentally sending values to fixtures with incompatible control layouts.

When a Pico base URL is set, each Group Edit change is sent to the Pico immediately. To use Group Edit browser-only, clear the Pico base URL first.

If the controller is currently scoped by a recalled scene, recalled palette, or Fan Out result, **Group Edit** uses that scope. In that case it shows only controls that are part of the active scope and exist on at least two selected fixtures. Editing a scoped control writes only to matching fixtures.

Use **Default all** or **Blackout all** to recall the stored default or blackout values for every fixture in the selected group.

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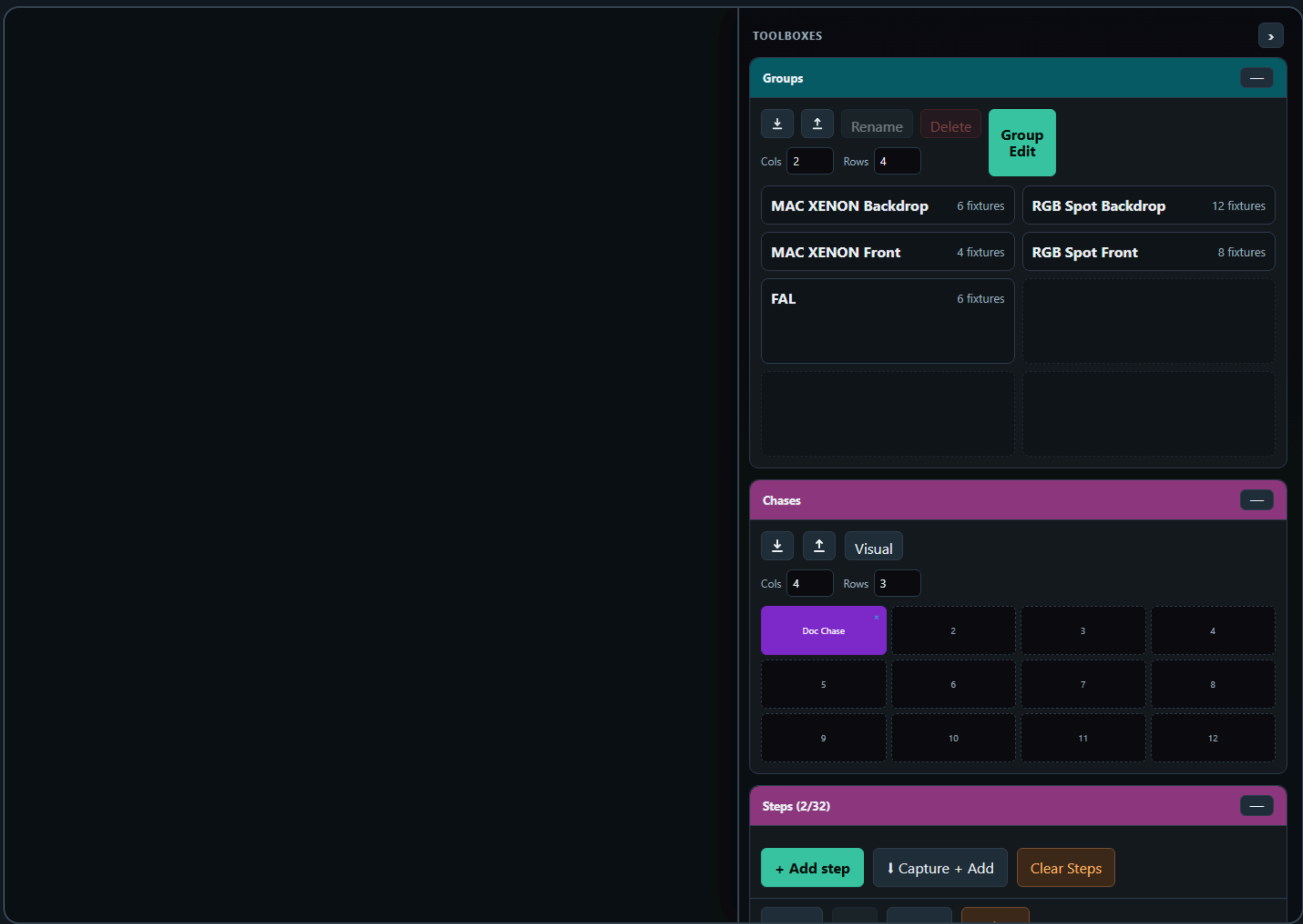
4. Chaser



The Chaser page creates step-based sequences. The main page stays focused on **Participating Controls** and **Edit Step**, while the repeated working tools sit in the **Toolboxes** sidebar.

The Chaser screenshot is captured with the important boxes visible on purpose: **Groups**, **Chases**, **Chase Steps**, and **Chase Playback**. Toolbox headers can be dragged up or down to reorder them in the sidebar. The order is shared with the other pages: if a page does not use one of the toolbox types, the next available toolbox moves up.

Basic Workflow

1. Open **Chaser**.
2. Select participating fixture controls.
3. Create steps with **Add step**, **Capture + Add**, or **Capture from FC**.
4. Set step duration and fade in **Edit Step**.
5. Store the chase in the **Chases** toolbox if you want quick recall.
6. Test timing in **Chase Playback**.

7. Click an empty Pico slot to upload the current chase to that slot.
8. Play the slot from the Pico.

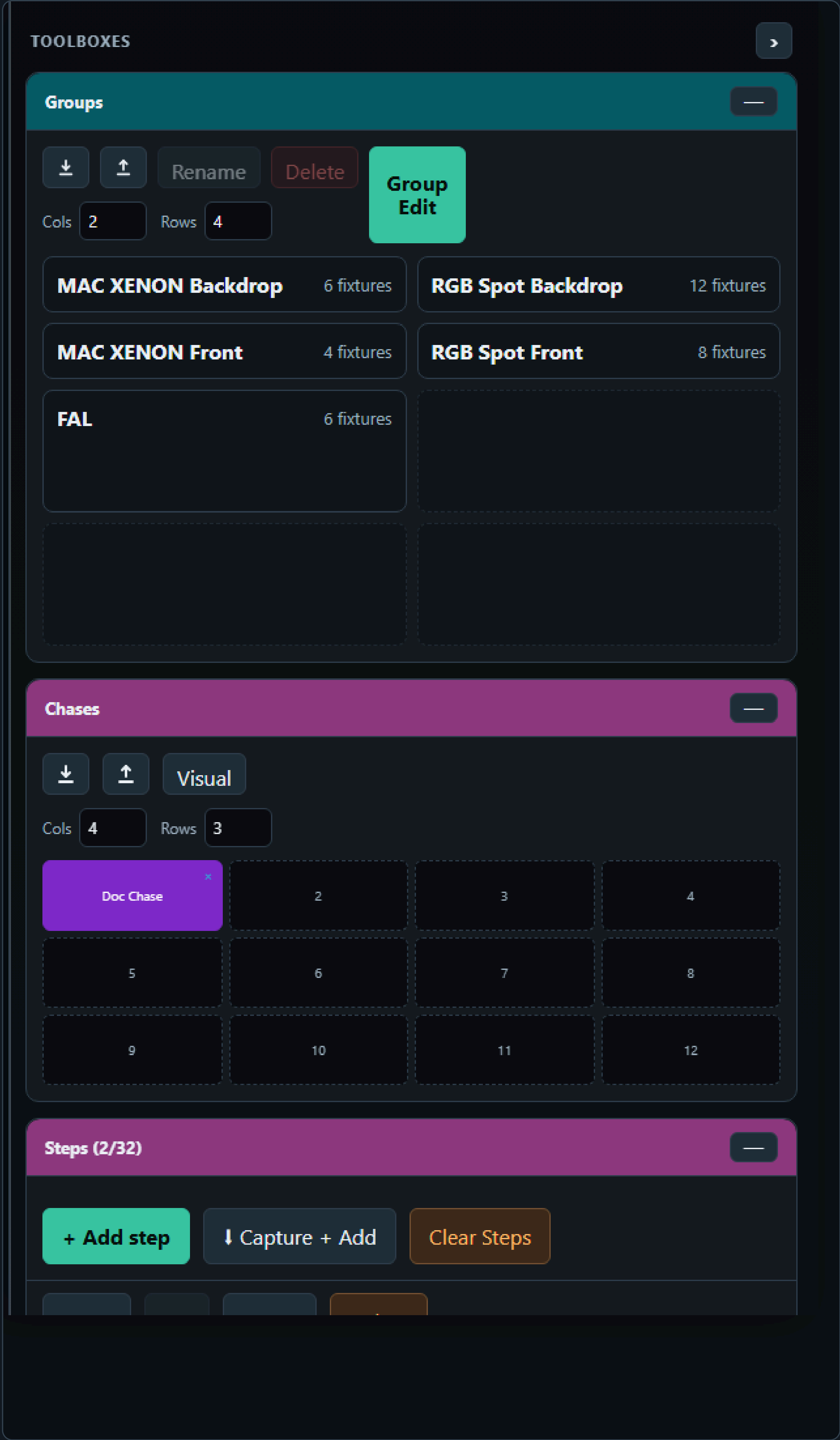
Toolbox Sidebar

Controller, Chaser, and Motion FX use a shared right-side toolbox sidebar on desktop-sized screens.

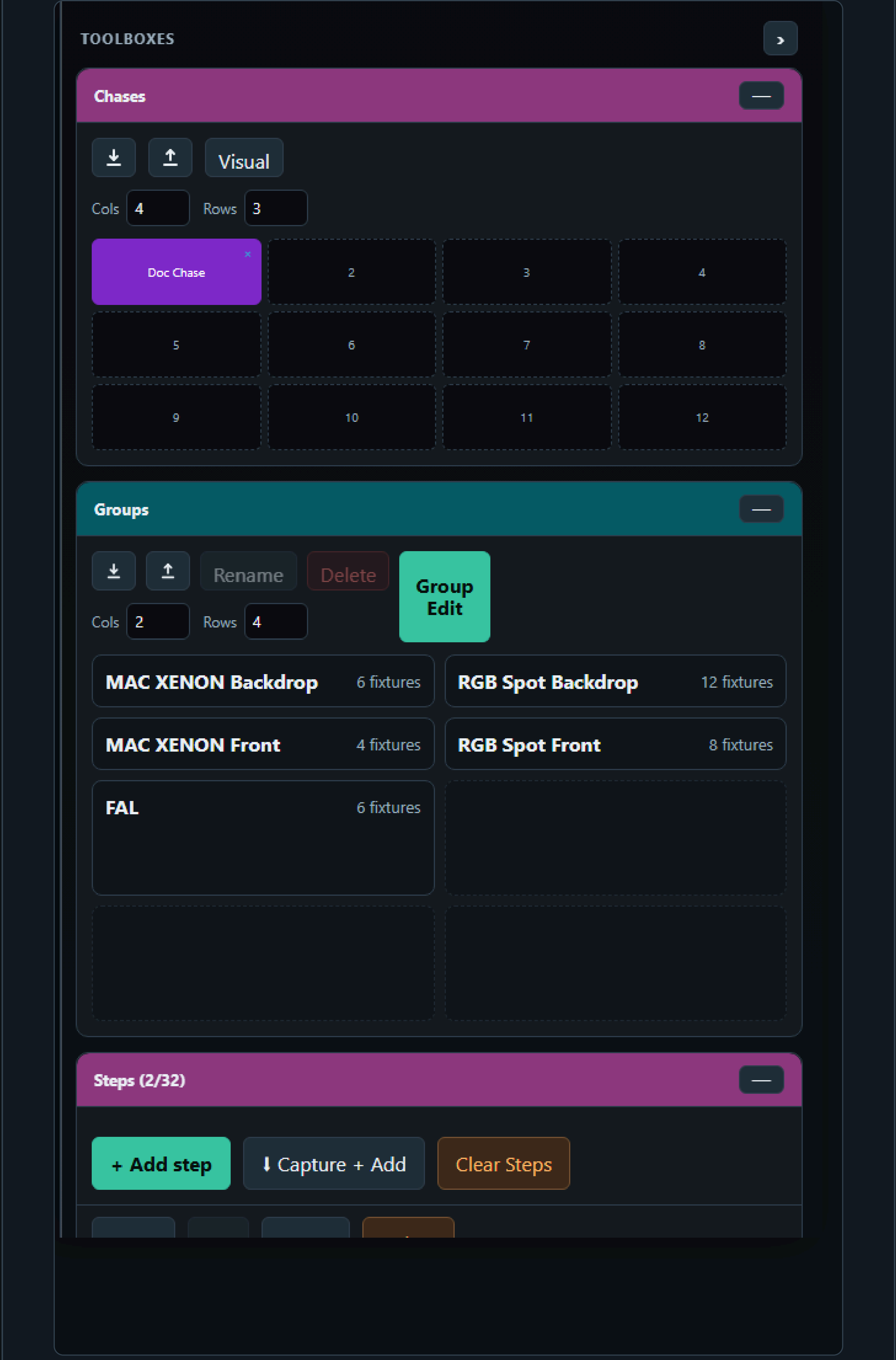
- Drag the vertical resize line on the left edge of the sidebar to change its width.
- The sidebar width is shared across all toolbox pages.
- Double-click the resize line to reset the default width.
- The page content and the toolbox sidebar scroll independently. The page scrollbar sits beside the toolbox separation line, while the toolbox scrollbar stays inside the toolbox sidebar.
- Use the arrow button in the Toolboxes header to collapse or reopen the whole sidebar. The collapsed state is shared across toolbox pages.
- Drag a toolbox by its colored header to reorder the sidebar. The toolbox body is not a drag handle, so slot clicks, sliders, and buttons are safe on touch screens.
- On narrow screens, the sidebar changes into a bottom toolbox drawer.

The Chaser page uses several toolboxes:

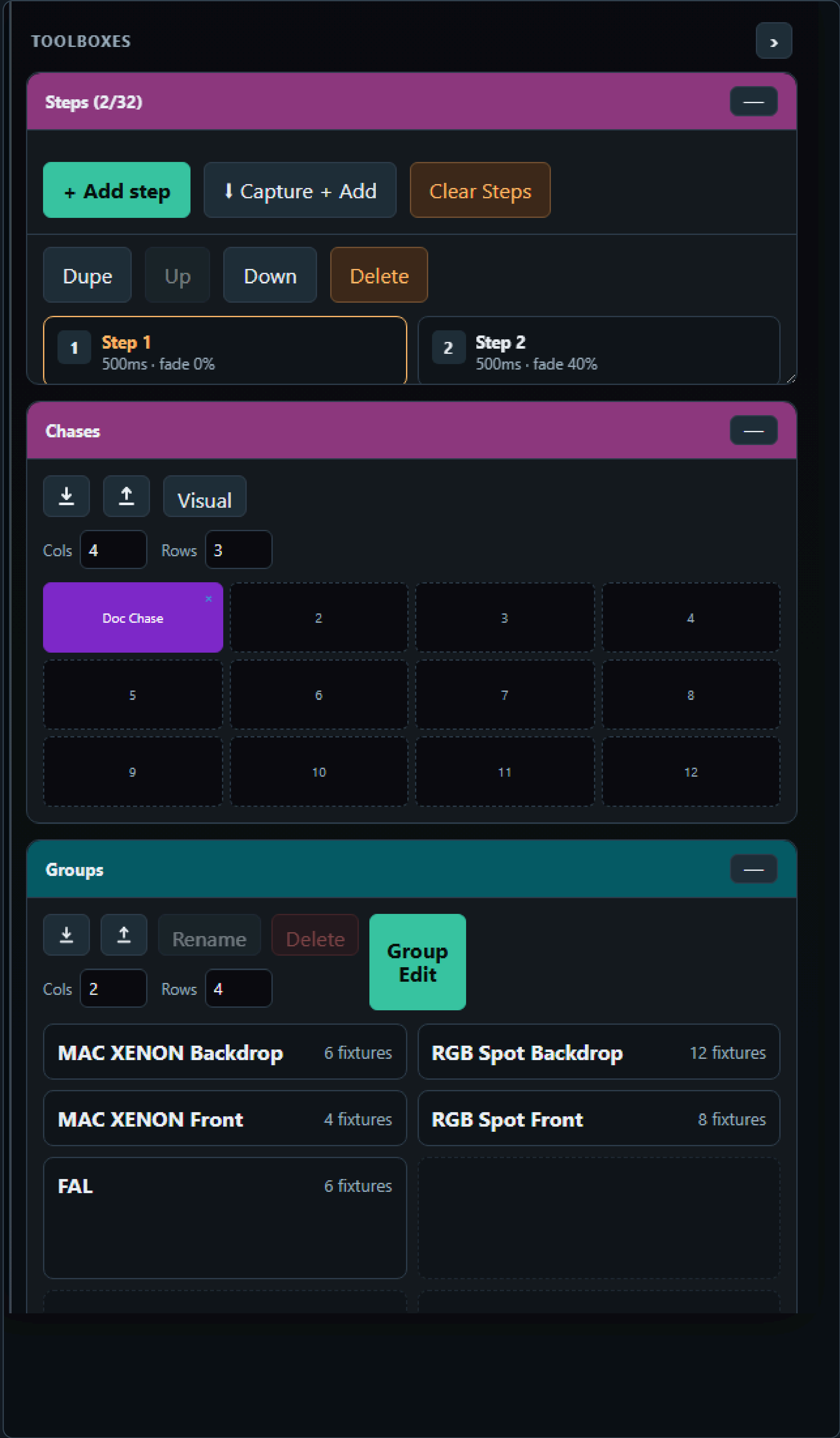
- **Groups** filters the fixture list by one or more saved fixture groups. It uses the cyan header, like the group tools on the controller page.


TOOLBOXES
Groups
Rename
Delete
Group Edit
Cols 2
Rows 4
MAC XENON Backdrop 6 fixtures
RGB Spot Backdrop 12 fixtures
MAC XENON Front 4 fixtures
RGB Spot Front 8 fixtures
FAL 6 fixtures
Chases
Visual
Cols 4
Rows 3
Doc Chase
2
3
4
5
6
7
8
9
10
11
12
Steps (2/32)
+ Add step
Capture + Add
Clear Steps

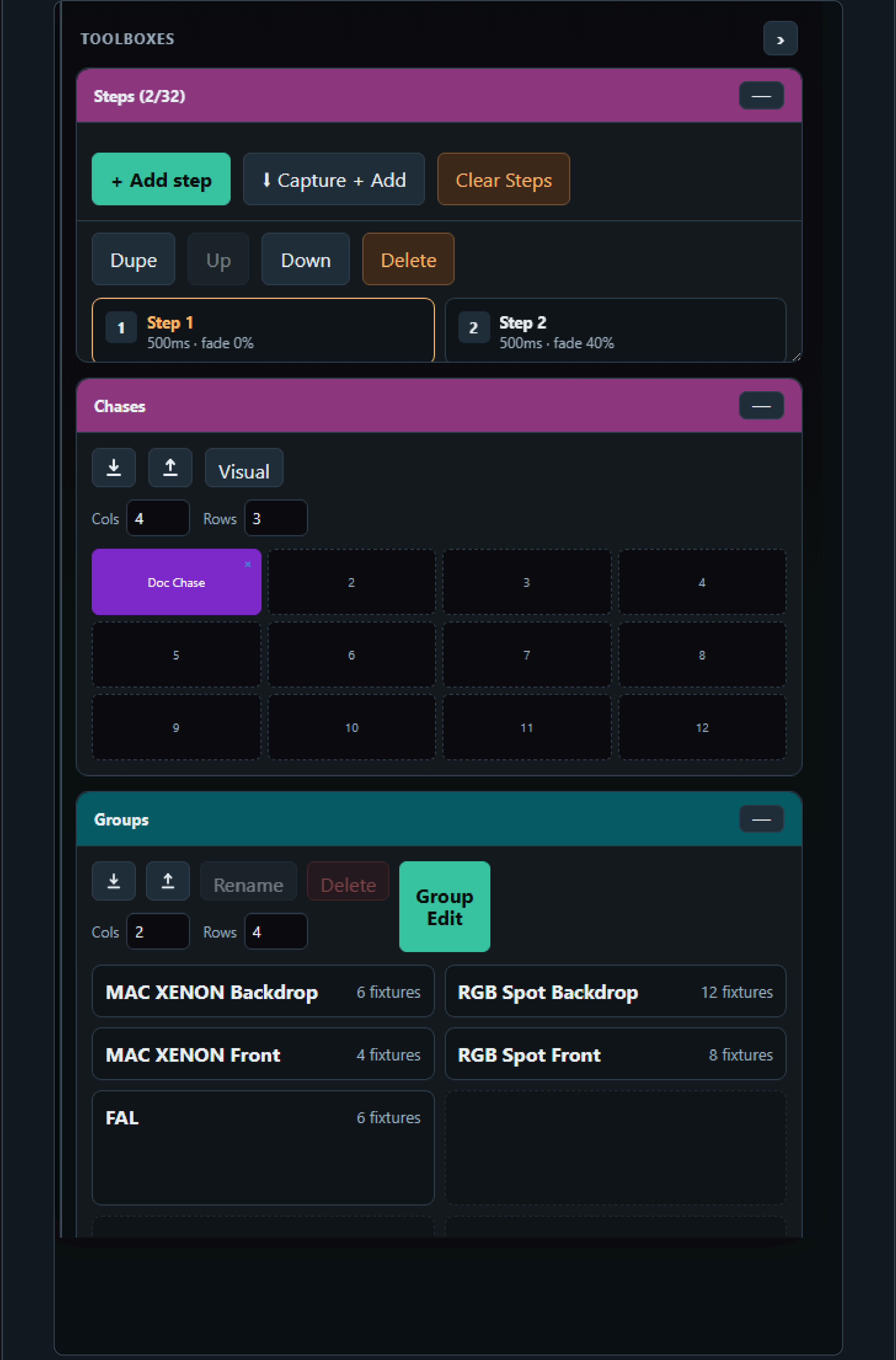
- **Chases** stores complete editable chases in a slot matrix. Clicking an empty slot saves the current chase. Clicking a filled slot loads that chase.
- The small top-left pencil icon on a filled **Chases** slot sets a background color and optional drawn/uploaded visual for that chase slot. **Default background** restores the standard slot color, and **No icon** removes the overlay image. This is only a label; loading a chase still uses the stored chase steps and playback settings.


TOOLBOXES
Chases
Visual
Cols 4
Rows 3
Doc Chase
2
3
4
5
6
7
8
9
10
11
12
Groups
Rename
Delete
Group Edit
Cols 2
Rows 4
MAC XENON Backdrop
6 fixtures
RGB Spot Backdrop
12 fixtures
MAC XENON Front
4 fixtures
RGB Spot Front
8 fixtures
FAL
6 fixtures
Steps (2/32)
+ Add step
⬇ Capture + Add
Clear Steps

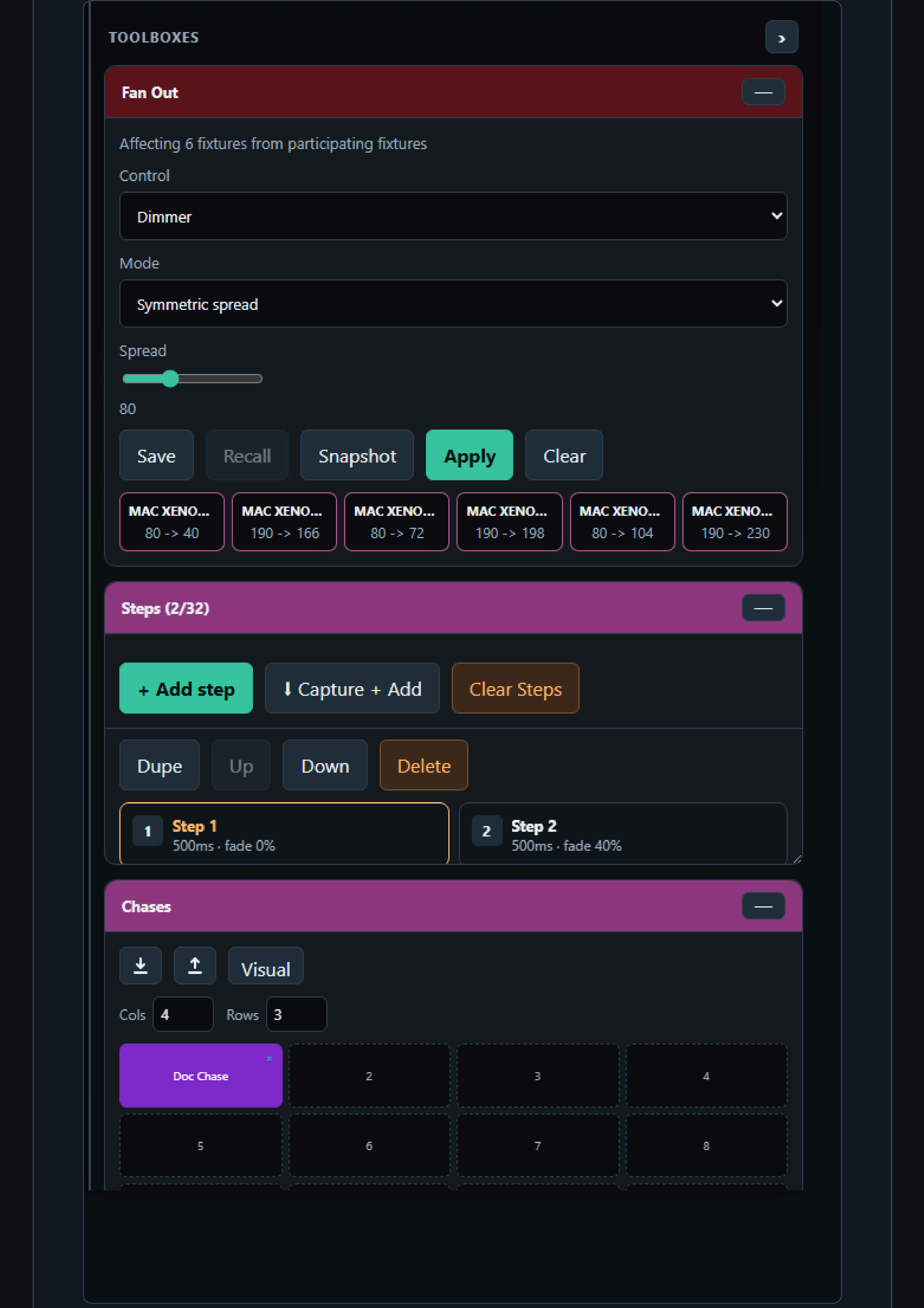
- **Palettes** stores and recalls reusable step fragments. Clicking an empty palette slot saves the currently selected step's fixture/control values to the shared palette JSON. Clicking a filled palette slot recalls compatible values into the selected step. **Merge** adds the selected step's values into an existing palette; if the palette scope differs, Chaser asks before changing it to **All controls**. The small top-left pencil icon edits a saved palette slot's visual appearance. When a Pico base URL is set, recalled palette values are also sent to the Pico.


TOOLBOXES
Steps (2/32)
+ Add step
⬇ Capture + Add
Clear Steps
Dupe
Up
Down
Delete
1
Step 1
500ms · fade 0%
2
Step 2
500ms · fade 40%
Chases
Visual
Cols 4
Rows 3
Doc Chase
2
3
4
5
6
7
8
9
10
11
12
Groups
Rename
Delete
Group Edit
Cols 2
Rows 4
MAC XENON Backdrop 6 fixtures
RGB Spot Backdrop 12 fixtures
MAC XENON Front 4 fixtures
RGB Spot Front 8 fixtures
FAL 6 fixtures

- **Chase Steps** contains the step list and step actions. Use it to add, capture, edit, duplicate, delete, and reorder steps. The box can be resized, and its top buttons remain visible while the list scrolls.
- **Chases**, **Chase Steps**, and **Chase Playback** share one toolbox color and collapse together. Use **-- all** on any of those boxes to collapse the whole color group, and **+ all** to reopen it.


TOOLBOXES
Steps (2/32)
+ Add step
Capture + Add
Clear Steps
Dupe
Up
Down
Delete
1
Step 1
500ms · fade 0%
2
Step 2
500ms · fade 40%
Chases
Visual
Cols 4
Rows 3
Doc Chase
2
3
4
5
6
7
8
9
10
11
12
Groups
Rename
Delete
Group Edit
Cols 2
Rows 4
MAC XENON Backdrop
6 fixtures
RGB Spot Backdrop
12 fixtures
MAC XENON Front
4 fixtures
RGB Spot Front
8 fixtures
FAL
6 fixtures

- **Fan Out** is a live step-shaping tool. It works on the currently selected step and writes directly into **Edit Step** as soon as you change the Fan Out mode, spread, or range values. There is no separate Snapshot or Apply action on the Chaser page.
- **Fan Out control selection** is filtered by the selected step. The control dropdown only shows compatible single-value controls that are actually part of the selected step's participating controls and exist on at least two fixtures. If one or more groups are selected, the same rule is applied inside the selected groups only.
- **Invert** reverses the fixture order for the Fan Out calculation. Use it when the spread shape is right but should run from the opposite side of the fixture row.
- **Fan Out base values** come from the values currently displayed in **Edit Step**. Selecting another step, loading another chase, capturing values, or using **Group Edit** refreshes the Fan Out base from the step values now shown on screen. This keeps spread calculations from drifting away from the edited step.
- **Clear** in the Chaser Fan Out toolbox resets the Fan Out shaping controls to neutral. It does not recall an older preset and it does not undo values that have already been written into the selected step.
- Editing **Edit Step**, using **Group Edit**, and changing **Fan Out** updates the selected step immediately. When a Pico base URL is set, those changed selected-step values are also sent to the Pico. Chase Playback sends the current chase continuously while it runs and therefore requires a Pico base URL.


TOOLBOXES
Fan Out
Affecting 6 fixtures from participating fixtures
Control
Dimmer
Mode
Symmetric spread
Spread
80
Save
Recall
Snapshot
Apply
Clear
MAC XENO... 80 -> 40
MAC XENO... 190 -> 166
MAC XENO... 80 -> 72
MAC XENO... 190 -> 198
MAC XENO... 80 -> 104
MAC XENO... 190 -> 230
Steps (2/32)
+ Add step
Capture + Add
Clear Steps
Dupe
Up
Down
Delete
1 Step 1 500ms · fade 0%
2 Step 2 500ms · fade 40%
Chases
Visual
Cols 4
Rows 3
Doc Chase
2
3
4
5
6
7
8

- **Chase Playback** runs the current chase from the browser for checking timing and fades before uploading to the Pico. The **Fade % (all steps)** field applies one fade value to every step immediately. Use **Edit Step > Fade %** when one step needs its own fade value.
- While Chase Playback runs, the currently playing step automatically becomes the selected step. This means **Edit Step** follows the chase visually and always shows the values of the last played step. Recalling a saved chase still selects Step 1 first, so playback starts from a predictable view.
- Selecting a step in **Chase Steps** or manually changing controls in **Edit Step** stops browser Chase Playback. This prevents playback from immediately overwriting the values you are trying to edit.

TOOLBOXES

›

Browser Playback

—

▶ Play | ■ Stop | ◀ Prev | Next ▶

Loop

Live preview

BPM

120

Beats/step

1

Tap tempo

Fade % (all steps)

0

Apply

Update rate (Hz)

25

Stopped

Fan Out

—

Affecting 6 fixtures from participating fixtures

Control

Dimmer

Mode

Symmetric spread

Spread

80

Save | Recall | Snapshot | Apply | Clear

MAC XENO...	MAC XENO...	MAC XENO...	MAC XENO...	MAC XENO...	MAC XENO...
80 -> 40	190 -> 166	80 -> 72	190 -> 198	80 -> 104	190 -> 230

Steps (2/32)

—

+ Add step | ⬇ Capture + Add | Clear Steps

Dupe | Up | Down | Delete

1 **Step 1**

2 **Step 2**

On page load, the Chaser working area starts with no steps selected. Use the **Chases** toolbox to recall a saved chase. Loading a chase from the **Chases** box updates the step list, selects Step 1, and rebuilds participating controls and the currently edited step together. If the chase contains steps, the participating controls are rebuilt from the values stored in the chase, so old fixture/group filters do not hide the controls used by that chase.

The collapse state, toolbox order, shared sidebar width, and the Chase Steps box size are stored by the server UI-state file, so the working layout survives reloads.

Chase Steps Toolbox Buttons

The **Chase Steps** toolbox uses shared action buttons instead of per-step buttons. Click a step card to select the step first; the selected step is highlighted and loaded into **Edit Step**.

- **Add step** creates a new selected step from the stored default values of the currently participating controls. If a fixture profile has no custom default for a control, Chaser uses the control type fallback.
- **Capture + Add** creates a new selected step only when Fixture Controller live values exist for at least one currently participating control. If no matching live values are available, no empty step is created.
- **Clear Steps** deletes all current working steps after confirmation. It does not delete saved Chases toolbox entries or Pico slot payloads.
- **Dupe** duplicates the selected step directly after itself and selects the copy.
- **Up** and **Down** move the selected step in the step order and keep the same step loaded in **Edit Step**.
- You can also drag a step card and drop it before or after another step card to reorder the chase. The selected step and the currently playing step stay attached to the same step data after the move.
- **Delete** removes the selected step. If another step remains, it becomes the selected step.

Participating Controls

Participating controls define which fixture controls belong to the chase. This keeps the chaser from editing unrelated channels.

For example, a dimmer chase might include only dimmer controls. A color chase might include only RGB or RGBWA controls.

If no group is selected, all patched fixtures are available. If one or more groups are selected in the Groups toolbox, only fixtures from those groups are shown while you are choosing the scope. Any direct change in **Participating Controls** then clears the group selection. This avoids mixing two different filters: after you press **All**, **None**, **Only**, **Add**, or tick an individual checkbox, the participating-control set becomes the source of truth.

The **Group control** dropdown is built from the fixtures currently visible in the participating-control scope, not from the controls that are already ticked. With no Groups filter, the dropdown stays based on all visible patched fixtures even after **Only** narrows the selected participating controls. With a Groups filter, the dropdown is based on the selected groups. When an edited step is active, the visible step fixtures define the dropdown scope.

All selects every valid participating control for all patched fixtures, clears the Groups filter, expands the participating fixture list, stops browser Chase Playback, clears the selected step, clears **Edit Step**, clears the Source fixture, and resets Fan Out bases. Existing steps in **Chase Steps** are not deleted. After **All**, use **Add step**, **Capture + Add**, or **Group Edit** to create or edit a step from the full participating-control scope.

Group Edit edits the current participating-control scope. It becomes available when the current scope contains at least one matching selected control on two or more participating fixtures. A step does not need to be selected first. If no step is selected, the first Group Edit value change creates a new step from the current participating controls and writes the edit into that step. If a step is selected, Group Edit edits that selected step.

The fixtures may use different profiles; the modal only shows matching controls that are actually selected as participating controls and exist on at least two fixtures. For example, if only Dimmer is selected, Group Edit opens with Dimmer only and applies it to every involved fixture that has a matching Dimmer control.

Group Edit uses the **Source** fixture from **Edit Step** as the reference value. Opening the modal does not automatically overwrite the other fixtures. Use **Apply source** on one control to copy that Source value to all matching participating fixtures for that control only. Use **All** at the lower-left of the modal to copy the Source values for every editable control in the modal.

Chaser Selection Rules

The Chaser page has two different selection modes: defining a new participating-control set, and editing or recalling an existing step. They intentionally behave differently.

Keep these rules as the contract for the Chaser page:

- **Participating Controls** define the fixture/control scope for new steps and for Group Edit.
- **Edit Step** shows the currently selected step, not a separate preview copy.
- If a saved chase is recalled, Step 1 is selected immediately and becomes the visible **Edit Step**.
- If **Chase Playback** is running, playback overrules manual step rendering: every played step automatically becomes the selected step and redraws **Participating Controls** plus **Edit Step**.
- During fades, **Edit Step** updates the existing sliders, readouts, XY dots, colors, and wheel highlights continuously from the interpolated playback values. The full card is only rebuilt when the played step changes.
- When playback stops or pauses, the last played step remains selected.
- Manual previous/next playback also selects the stepped-to chase step and redraws **Edit Step**.
- Group filters are only temporary scope builders. Direct Participating Controls changes clear the group filter.
- **All** clears the selected step/edit context but keeps the existing step list unchanged.
- **Group Edit** does not require a selected step. If no step is selected, the first Group Edit value change creates a new step from the current participating controls. If a step is selected, Group Edit edits that step.

When you define participating controls manually:

- If no group is selected, the Participating Controls panel shows all patched fixtures.
- If one or more groups are selected, the panel is temporarily filtered to the fixtures in those groups.
- **All** selects every currently visible control and clears the group selection.
- **None** clears the participating-control selection, collapses the fixture list, and clears the group selection.
- The control dropdown lists controls from the selected groups while a group filter is active. If no group is selected, it lists controls from the current participating-control scope.
- The control dropdown plus **Only** selects one matching control type from the current scope, clears all other controls, and then clears the group selection.
- The control dropdown plus **Add** adds one matching control type from the current scope without clearing existing participating controls, then clears the group selection.
- Ticking or unticking an individual participating-control checkbox also clears the group selection.

When you click a step in the **Chase Steps** toolbox:

- The step values are checked against the current fixture setup.
- Invalid fixture/control references are removed from the edited step.
- The Participating Controls panel is rebuilt from the chase values.
- If no group is selected, only fixtures and controls that are actually stored in the selected step are shown, and **Group Edit** becomes step-dependent.
- If a group is selected, the selected step is filtered to that group and **Group Edit** uses the grouped fixtures as its edit scope.
- The Edit Step card shows the same scoped fixture/control set, so editing Step 2 cannot accidentally show or edit unrelated controls from another group or old filter.

When you click a saved chase in the **Chases** toolbox:

- The selected Groups filter is cleared first.
- The saved steps, playback settings, and Chase Playback settings are loaded.
- The first step becomes the edited step.
- Participating Controls and Edit Step are rebuilt from that loaded step.
- If the saved chase was made before fixtures were repatched, the page tries to remap old fixture IDs to the current setup by matching the same fixture profile in DMX start-address order.
- If no valid step values can be found, the page falls back to the saved participating-control map stored with the chase.

This means group selection is a tool for building, filtering, and group-editing a step. Loading a saved chase clears the group filter because the recalled chase data becomes the source of truth. After recall, **Group Edit** can still become available without a selected group because it uses the fixtures participating in the selected step.

Create Chase Steps

A chase step stores values for the selected **Participating Controls**. It does not store every patched fixture automatically.

To create a step manually:

1. Select the controls that should participate in **Participating Controls**.
2. In the **Chase Steps** toolbox, click **Add step**.
3. Click the new step in the **Chase Steps** list.
4. In **Edit Step**, set **Label**, **Duration (ms)**, and **Fade %**.
5. Adjust the controls shown in **Edit Step**. Those control values are written into the selected step immediately. When a Pico base URL is set, they are also sent to the Pico.
6. Click **Apply** after changing the label, duration, or fade.

Add step starts from the stored default values of the selected participating controls. If a fixture profile has no custom default for a control, Chaser uses the control type fallback: centered pan/tilt, zero for sliders, wheels, and color channels.

After **Add step** or **Capture + Add**, the first fixture in the new step is automatically marked as the **Source** fixture in **Edit Step**. Click another fixture card in **Edit Step** to make it the Source fixture. Group Edit uses the Source fixture's current step values as its starting values before writing changes to the matching participating fixtures.

Use the **Palettes** toolbox when the selected step should become a reusable fragment. Empty palette slots save exactly the selected step's stored values, not the whole chase and not all DMX channels. Filled palette slots merge compatible palette values into the selected step.

To capture a new step from the Fixture Controller:

1. Open **Fixture Controller** in another browser tab or window.
2. Build the look there by moving controls, recalling a scene, or recalling a palette.

3. Return to **Chaser**.
4. Make sure **Participating Controls** contains the controls you want to capture.
5. In the **Chase Steps** toolbox, click **Capture + Add**.

Capture + Add creates a new step and copies the current Fixture Controller live values for the selected participating controls.

To capture into an existing step:

1. Click the step in the **Chase Steps** list.
2. Build the wanted look in **Fixture Controller**.
3. Return to **Chaser**.
4. In **Edit Step**, click **Capture from FC**.

Capture from FC updates the selected step with the current Fixture Controller live values for the selected participating controls.

Capture From Fixture Controller

Use **Capture from FC** or **Capture + Add** to read the current Fixture Controller live values and use them as chase step values.

This is useful when you want to build a chase visually:

1. Create a look on the Fixture Controller.
2. Capture it into the Chaser.
3. Change the look.
4. Capture the next step.

If Chaser reports that no Fixture Controller values are available, open the Fixture Controller page and move a control or recall a scene first. Capture only reads controls that are selected as participating controls, so unselected controls are ignored.

Pico Slot Playback

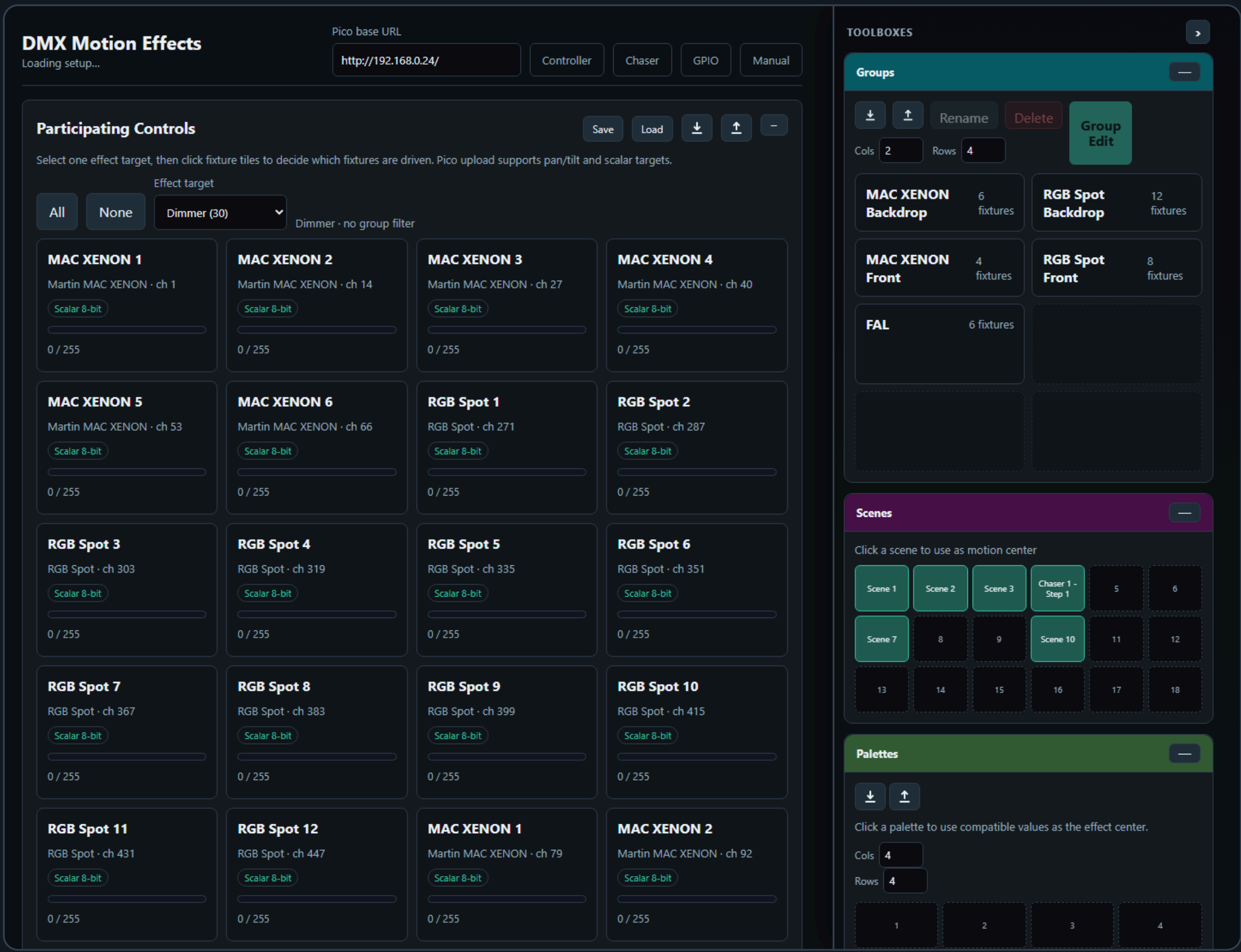
Chasers can be uploaded to 32 Pico slots. Each slot can run on the Pico without the browser staying open. The Pico slot strip is the upload target: click an empty slot to upload the current chase. Click a loaded slot once to select it for play, pause, resume, speed, or stop. Click the selected loaded slot again if you want to replace it with the current chase; the page asks before overwriting.

Supported playback options:

- Single run
- Loop
- Loop N times
- Forward or reverse direction
- Speed multiplier
- Pause and resume

Each chaser slot supports up to 32 steps.

5. Motion FX



Motion FX creates continuous effects for one selected effect target at a time. It can drive a combined pan/tilt target, or one scalar DMX target such as dimmer, prism, gobo, zoom, or iris.

Supported effects include:

- Pan/tilt targets: Circle, Figure-8, Pan swing, Tilt swing
- Scalar targets: Sine, Pulse

Pan/tilt is treated as one combined two-axis target. Pan/tilt effects are relative to the current scene position, so the effect moves around the position that was last written into the Pico base buffer. Scalar controls are one-axis targets and use their displayed center value plus the **Pan / scalar amplitude** as the effect depth.

The **Participating Controls** panel uses an **Effect target** dropdown. One effect can only target one control type at a time: either pan/tilt, or one scalar control type. Changing the effect target rebuilds the fixture matrix and prevents mixed targets such as dimmer plus gobo plus pan/tilt in one effect.

The fixture matrix is a selection and preview surface:

- Click a fixture tile to include or exclude it from the effect.
- Use **All** to clear any group filter and enable every fixture for the current target.
- Use **None** to disable every visible fixture for the current target.
- Selecting groups applies an additional filter and hides fixtures outside the selected groups.
- Pan/tilt targets show a small XY plot with the current position.
- Scalar targets show a small value bar with the current value.

Center values come from the current base buffer or from recalling a scene as the motion center. Phase spread, amplitude, BPM, and effect shape are set in the **Effect Parameters** toolbox.

The **Effect** dropdown is target-aware. It only shows effects that make sense for the selected **Effect target**.

The same target rules are used for Pico upload. Pan/tilt and scalar effects can be uploaded to one of the Pico Motion slots, and the Pico reads the effect center from its base buffer while playing.

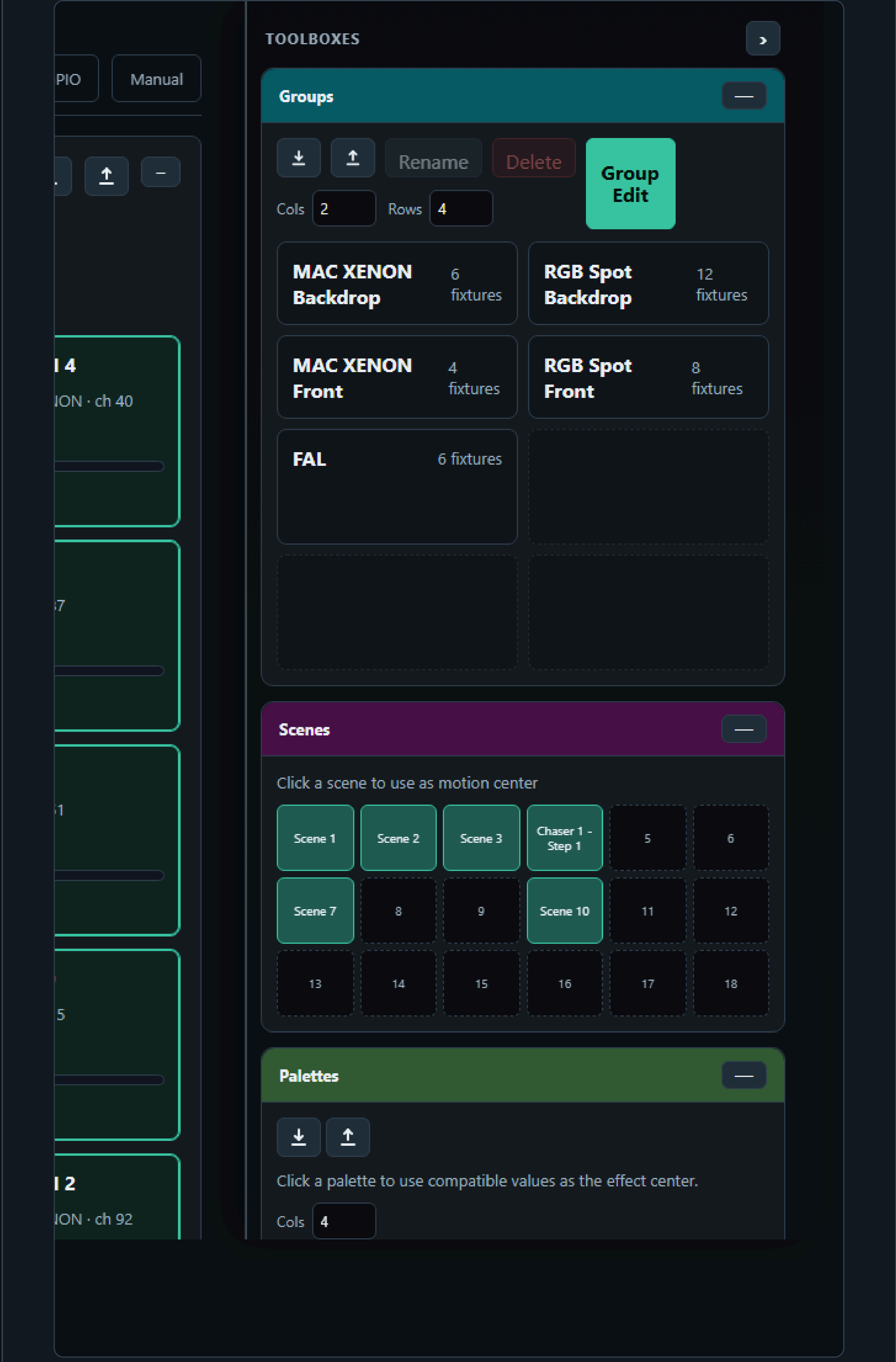
The Motion FX page also includes the shared **Palettes** toolbox. Clicking a palette recalls any values that are compatible with Motion FX and uses them as the current effect center. For example, a position palette can set pan/tilt centers, while a dimmer or beam palette can set scalar centers. Palette creation and visual editing still happen on the Fixture Controller; Motion recalls, imports, and exports the shared palette JSON.

The **Effects** toolbox stores reusable effect recipes. Click an empty effect slot to save the selected Effect target, participating fixtures, effect type, BPM, amplitudes, spread, and phase offsets. Effects do not store the current center/base values, so the same saved effect can be reused with different scene or palette centers. Clicking a saved effect recalls the recipe without sending DMX and without uploading to a Pico slot.

Effect Parameters and **Effects** share one toolbox color and collapse together. Use **-- all** on either box to collapse the whole effect group, and **+ all** to reopen it.

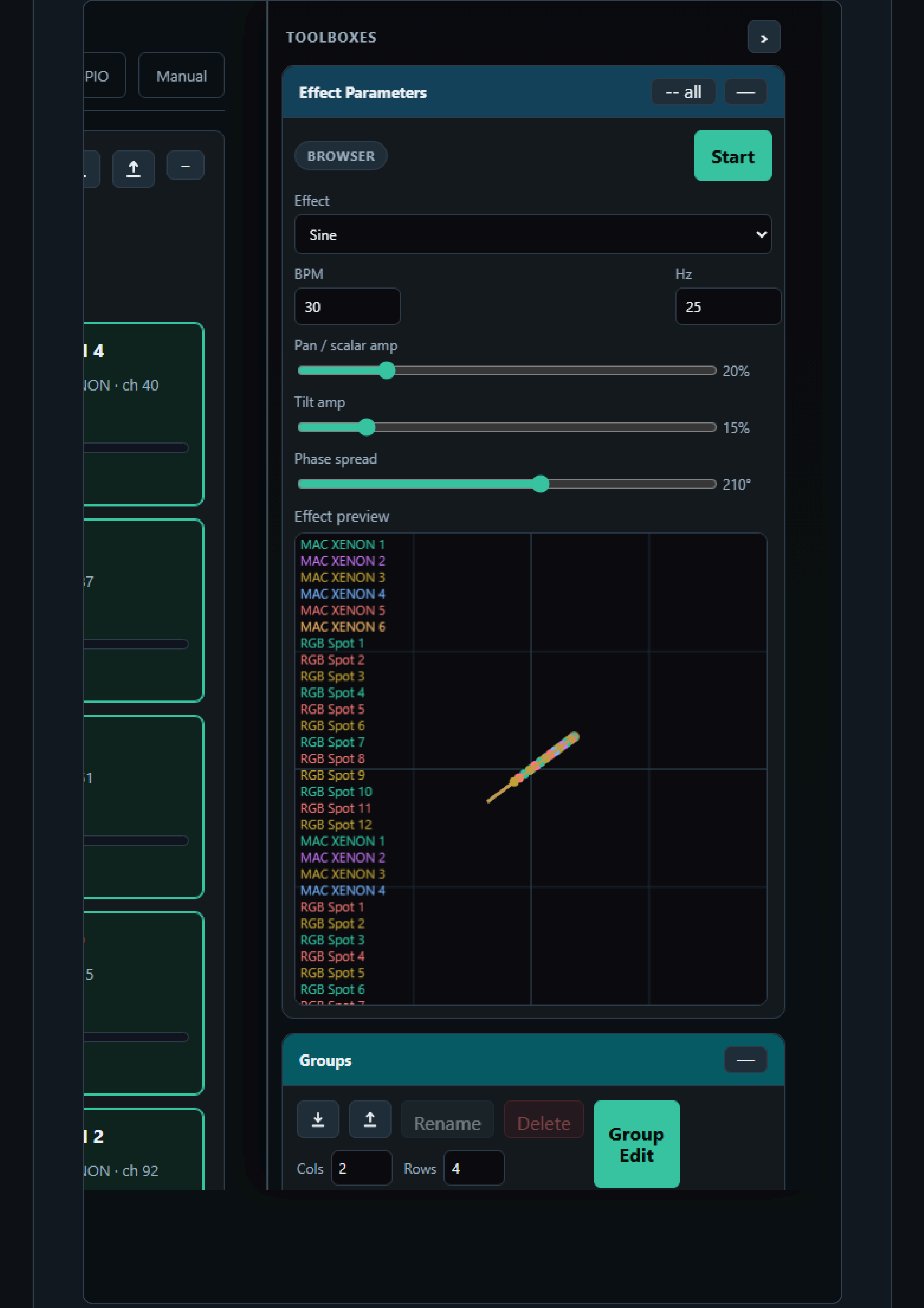
Motion FX Toolboxes

The Motion FX page uses five toolboxes in the shared sidebar.



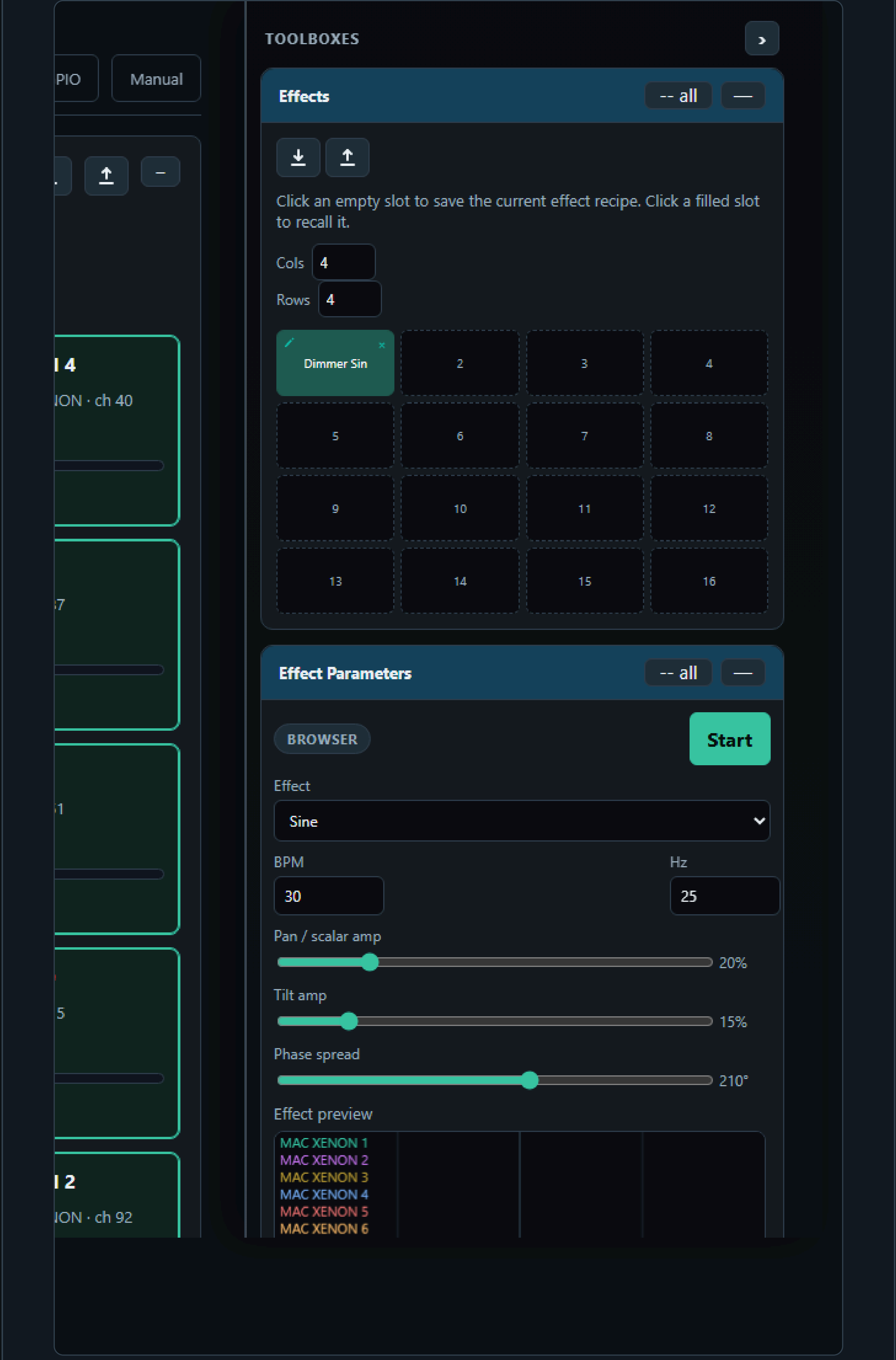
TOOLBOXES
PIO
Manual
Groups
Rename
Delete
Group Edit
Cols 2
Rows 4
MAC XENON Backdrop
6 fixtures
RGB Spot Backdrop
12 fixtures
MAC XENON Front
4 fixtures
RGB Spot Front
8 fixtures
FAL
6 fixtures
Scenes
Click a scene to use as motion center
Scene 1
Scene 2
Scene 3
Chaser 1 - Step 1
5
6
Scene 7
8
9
Scene 10
11
12
13
14
15
16
17
18
Palettes
Click a palette to use compatible values as the effect center.
Cols 4

Groups filters the fixture matrix for the selected effect target. Pressing **All** clears the group filter and enables every fixture available for the current target. **Group Edit** edits compatible selected target controls across two or more fixtures.

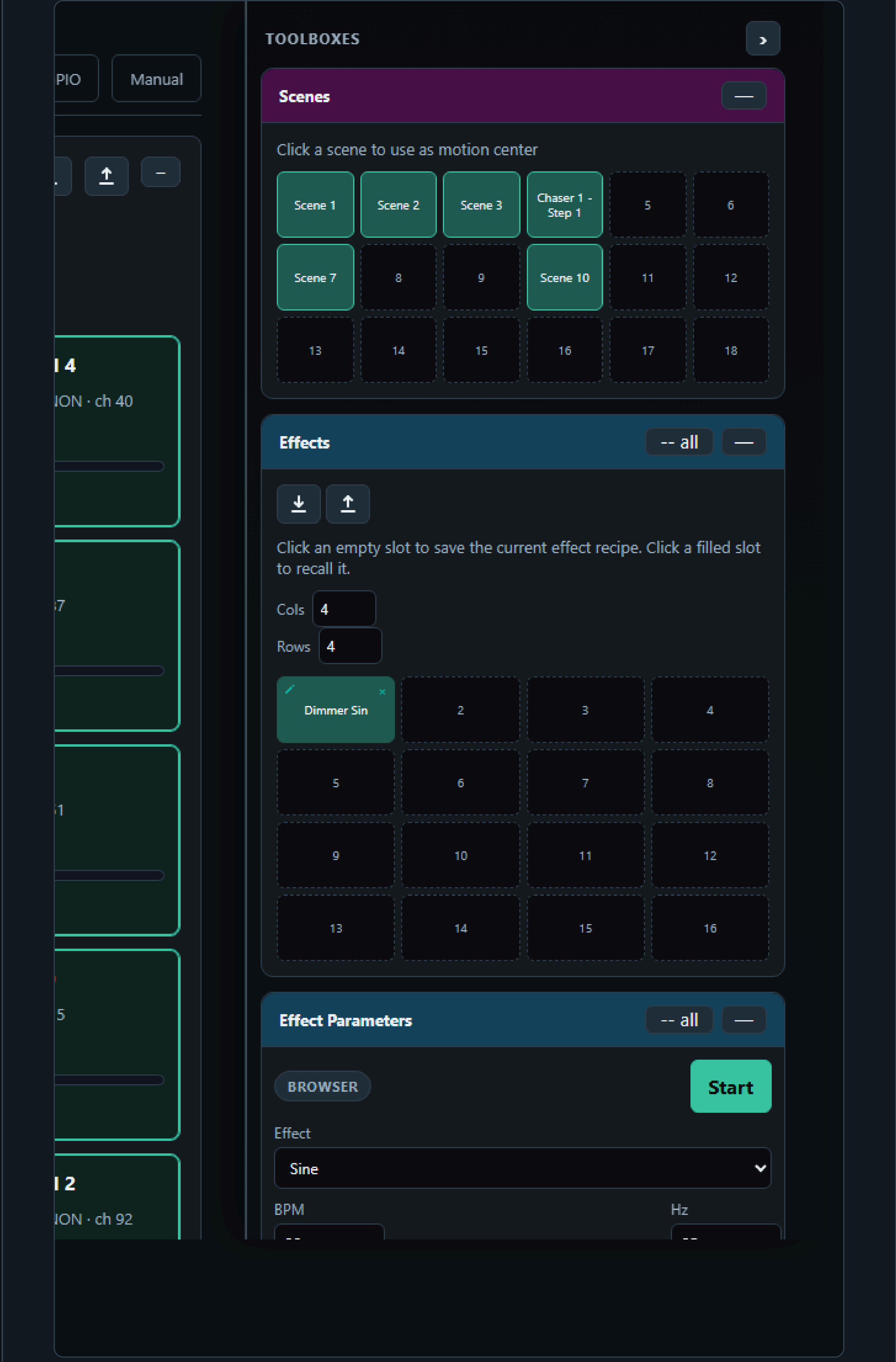


PIO
Manual
TOOLBOXES
Effect Parameters
-- all
BROWSER
Start
Effect
Sine
BPM
30
Hz
25
Pan / scalar amp
20%
Tilt amp
15%
Phase spread
210°
Effect preview
MAC XENON 1
MAC XENON 2
MAC XENON 3
MAC XENON 4
MAC XENON 5
MAC XENON 6
RGB Spot 1
RGB Spot 2
RGB Spot 3
RGB Spot 4
RGB Spot 5
RGB Spot 6
RGB Spot 7
RGB Spot 8
RGB Spot 9
RGB Spot 10
RGB Spot 11
RGB Spot 12
MAC XENON 1
MAC XENON 2
MAC XENON 3
MAC XENON 4
RGB Spot 1
RGB Spot 2
RGB Spot 3
RGB Spot 4
RGB Spot 5
RGB Spot 6
Groups
Rename
Delete
Group Edit
Cols
2
Rows
4

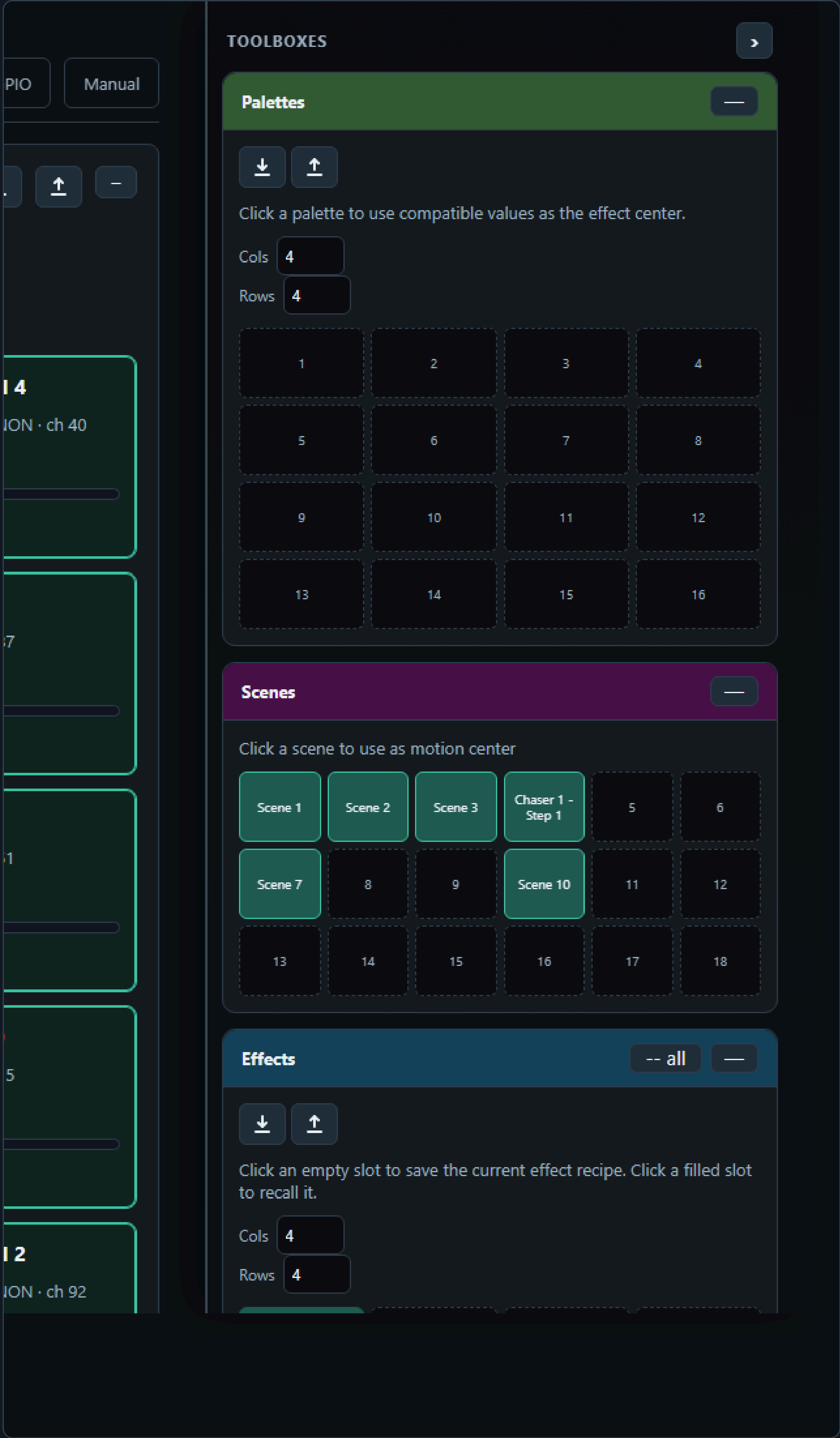
Effect Parameters is the live effect editor. It contains the target-aware effect dropdown, BPM, amplitude, phase spread, browser preview controls, and the Pico slot upload/play controls.


PIO
Manual
TOOLBOXES
Effects
-- all
Click an empty slot to save the current effect recipe. Click a filled slot to recall it.
Cols 4
Rows 4
Dimmer Sin
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
Effect Parameters
-- all
BROWSER
Start
Effect
Sine
BPM
30
Hz
25
Pan / scalar amp
20%
Tilt amp
15%
Phase spread
210°
Effect preview
MAC XENON 1
MAC XENON 2
MAC XENON 3
MAC XENON 4
MAC XENON 5
MAC XENON 6
ON · ch 40
ON · ch 92

Effects stores reusable effect recipes. It saves the selected target, fixture participation, effect type, BPM, amplitude, spread, and phase offsets. It does not store fixed center values.


TOOLBOXES
PIO
Manual
Scenes
Click a scene to use as motion center
Scene 1
Scene 2
Scene 3
Chaser 1 - Step 1
5
6
Scene 7
8
9
Scene 10
11
12
13
14
15
16
17
18
l 4
ION · ch 40
7
1
5
l 2
ION · ch 92
Effects
-- all
Click an empty slot to save the current effect recipe. Click a filled slot to recall it.
Cols 4
Rows 4
Dimmer Sin
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
Effect Parameters
-- all
BROWSER
Start
Effect
Sine
BPM
Hz

Scenes is read-only on Motion FX. Recalling a scene updates the motion center/base values used by effects. When a Pico base URL is set, the stored values are also sent to the Pico.


PIO
Manual
TOOLBOXES
Palettes
Click a palette to use compatible values as the effect center.
Cols 4
Rows 4
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
Scenes
Click a scene to use as motion center
Scene 1
Scene 2
Scene 3
Chaser 1 - Step 1
5
6
Scene 7
8
9
Scene 10
11
12
13
14
15
16
17
18
Effects
-- all
Click an empty slot to save the current effect recipe. Click a filled slot to recall it.
Cols 4
Rows 4
l 4
ION · ch 40
l 2
ION · ch 92

Palettes recalls compatible palette values into Motion FX. A position palette can set pan/tilt centers, while color, dimmer, or beam palettes can set scalar centers when the selected effect target matches.

Pico Motion Slots

The Pico Motion slot upload now uses the same selected **Effect target** as the browser page.

- If the target is pan/tilt, the uploaded slot stores pan and tilt channel addresses and plays two-axis effects.
- If the target is scalar, the uploaded slot stores that one control's DMX channel address and plays one-axis effects such as sine or pulse.
- Click an empty Pico slot to upload the current effect to that slot.
- Click a loaded slot once to select it for start, stop, or BPM changes.
- Click the selected loaded slot again to replace it with the current effect; the page asks before overwriting.
- Slots store channel mappings, BPM, amplitude, spread, effect type, and target phase. They do not store fixed center values.
- The center value is read from the Pico base buffer during playback. This means a scene recall or live controller change can define the center before the slot starts.
- Up to 64 Motion FX slots can be loaded on the Pico.

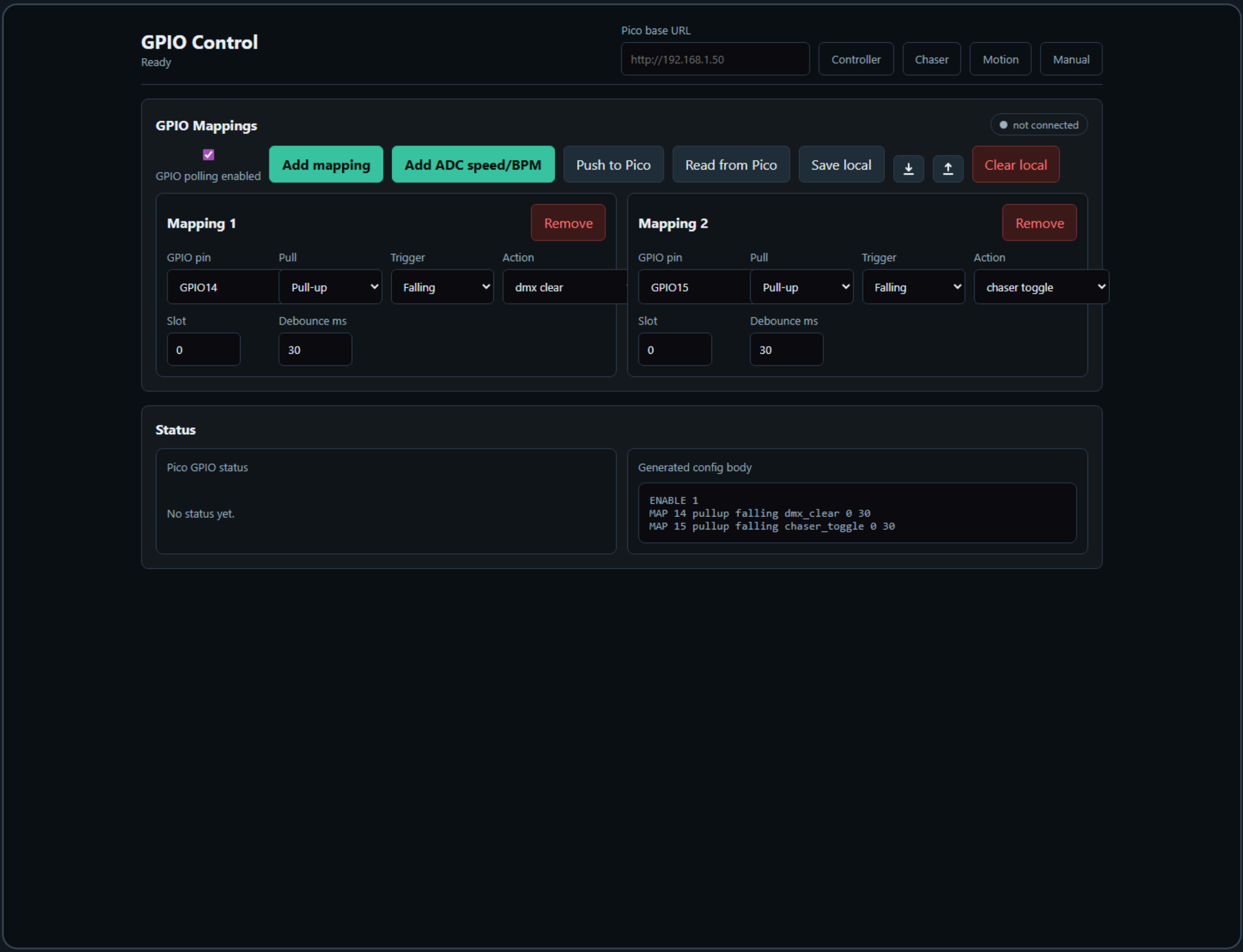
For scalar targets, set the current value first, then upload/start the slot. For example, set a dimmer to its desired base brightness and use Sine if you want the Pico to pulse above and below that base value.

Recommended Workflow

1. Use the Fixture Controller to position the fixtures.
2. Save or recall a scene.
3. Open Motion FX.
4. Select one **Effect target**.
5. Set BPM, effect shape, amplitude, and spread in the **Effect Parameters** toolbox.
6. Optionally recall a palette from the **Palettes** toolbox to set the center for the selected target.
7. Optionally save or recall the recipe from the **Effects** toolbox.
8. Click an empty Pico slot to upload the effect, then start the slot when you want autonomous playback without browser timing jitter.
9. Use browser playback from the **Effect Parameters** toolbox when you want quick live testing before uploading.

The Motion FX page also has a read-only scene toolbox. Clicking a scene updates the effect center. When a Pico base URL is set, it also sends the position to the Pico.

6. GPIO Control



GPIO Control maps physical Pico inputs to lighting actions.

GPIO Control does not use the shared toolbox sidebar. Its setup is kept in normal page panels because GPIO mapping is configuration work, not a live fixture/scene/palette workflow.

Digital GPIO pins can trigger:

- DMX clear
- Output-only clear
- Stop all
- Chaser play, stop, toggle, pause, resume, pause toggle
- Chaser tap tempo
- Motion start, stop, toggle
- Motion tap tempo

ADC pins can control:

- Chaser speed multiplier
- Motion FX BPM

Only GPIO26, GPIO27, and GPIO28 support ADC input on the Pico 2 W.

Add a Digital Button Mapping

1. Open **GPIO Control**.

2. Click **Add mapping**.
3. Select a free GPIO pin.
4. Choose pull mode and trigger edge.
5. Choose the action.
6. Set the slot number if the action needs one.
7. Upload the config to the Pico.

The page disables reserved pins and pins already used by other mappings.

Add an ADC Mapping

1. Add an ADC mapping.
2. Select GPIO26, GPIO27, or GPIO28.
3. Choose `chaser_speed` or `motion_bpm`.
4. Set the target slot.
5. Set the min and max value.
6. Upload the config.

ADC values are filtered on the firmware side with a short mean filter to reduce ripple.

Tap Tempo

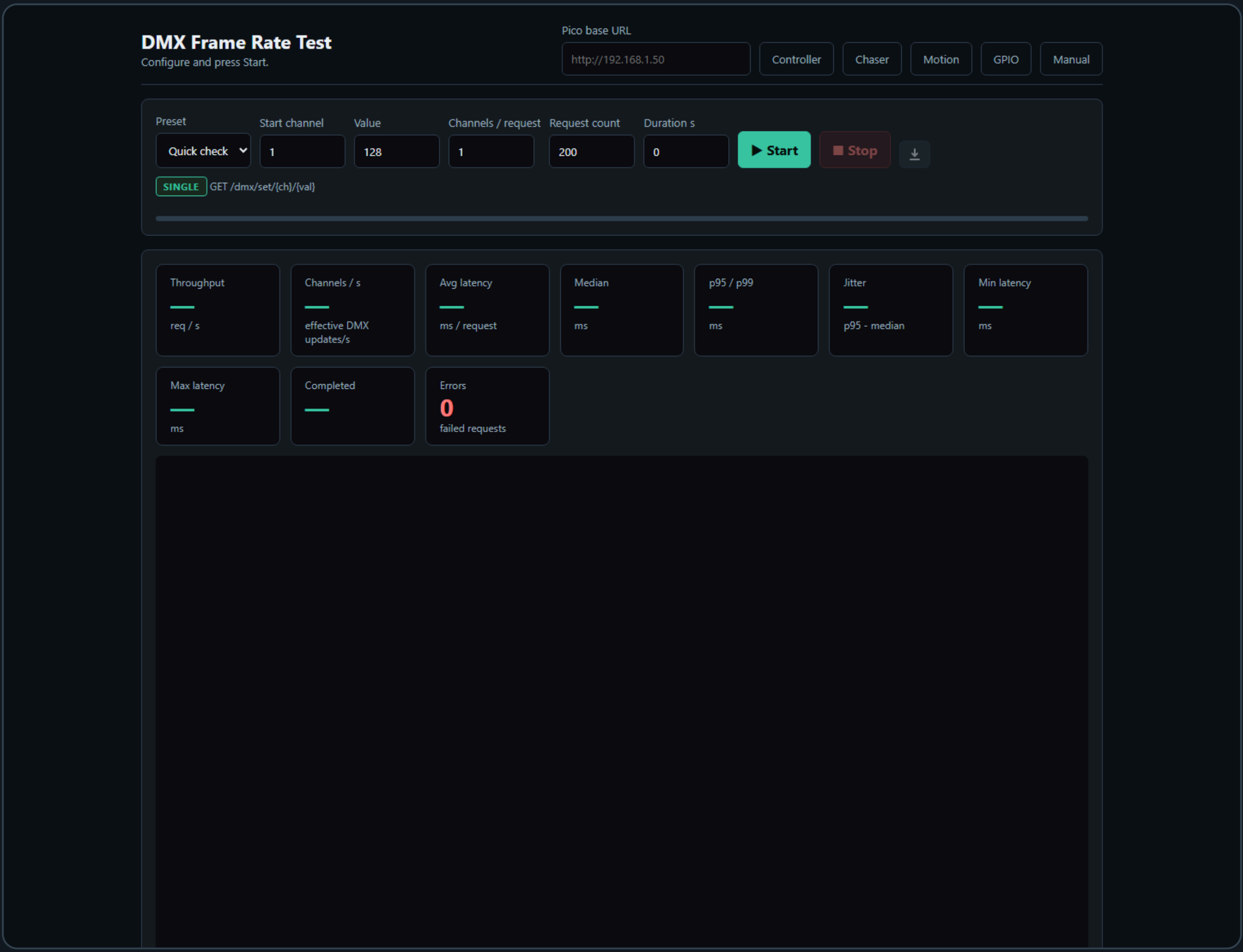
Tap actions use the time between button presses.

The beat divider can be set to:

- 1 beat
- 1/2 beat
- 1/4 beat
- 1/8 beat
- 1/16 beat

The firmware ignores very long gaps between taps so stopping for a while does not create an extremely slow tempo when tapping resumes.

7. Benchmark



The Benchmark page tests Pico HTTP and DMX update performance.

Benchmark does not use the shared toolbox sidebar. It is a developer/test page for measuring request performance and is documented as a single page panel.

Use it to compare:

- Single-channel updates
- Scene-sized batch updates
- Large batch updates
- Longer soak tests

The result panel shows:

- Throughput
- Effective DMX channel updates per second
- Average latency
- Median latency
- p95 and p99 latency
- Jitter
- Errors

Use **Export CSV** to save results for later comparison.

8. Backup and Import

Most setup data is stored as JSON files on the XAMPP server in the `data/` folder.

The UI also offers export/import buttons for:

- Fixture setup
- Groups
- Scenes
- Palettes
- Chaser setup
- Motion FX setup
- GPIO setup

Use JSON export before large changes so a known-good setup can be restored later.

---

9. Clear Functions

There are two different clear actions:

Action	What it clears	When to use
DMX clear	Live DMX output and the motion base buffer	Full reset of output and base position
Output-only clear	Live DMX output only	Black out output while keeping the stored motion center

Use output-only clear when you want Motion FX to resume around the same stored center after the blackout.

Troubleshooting

The UI does not control the Pico

- Check that the Pico is powered and connected to WiFi.
- Open the Pico URL directly in a browser, for example `http://192.168.0.24/`.
- Make sure the base URL in the UI ends with `/`.
- Make sure the Fixture Controller has the correct Pico base URL.

Chaser capture does not contain the expected values

- Move or recall the values on the Fixture Controller first.
- Make sure the participating controls include the controls you want to capture.
- Save or reload the Fixture Controller setup if fixtures were changed recently.

Motion FX moves around the wrong center

- Recall the desired scene first.
- On the Motion FX page, click the same scene in the scene toolbox.
- Then start the motion slot.

GPIO mapping does not react

- Check `/gpio/status` or the status panel on the GPIO page.
- Confirm that the selected pin is not reserved or already used.
- Check pull mode and trigger edge.
- For ADC, use only GPIO26, GPIO27, or GPIO28.

A chaser has more than 32 steps

The firmware supports 32 steps per chaser slot. Keep the chase within this limit before uploading to the Pico.